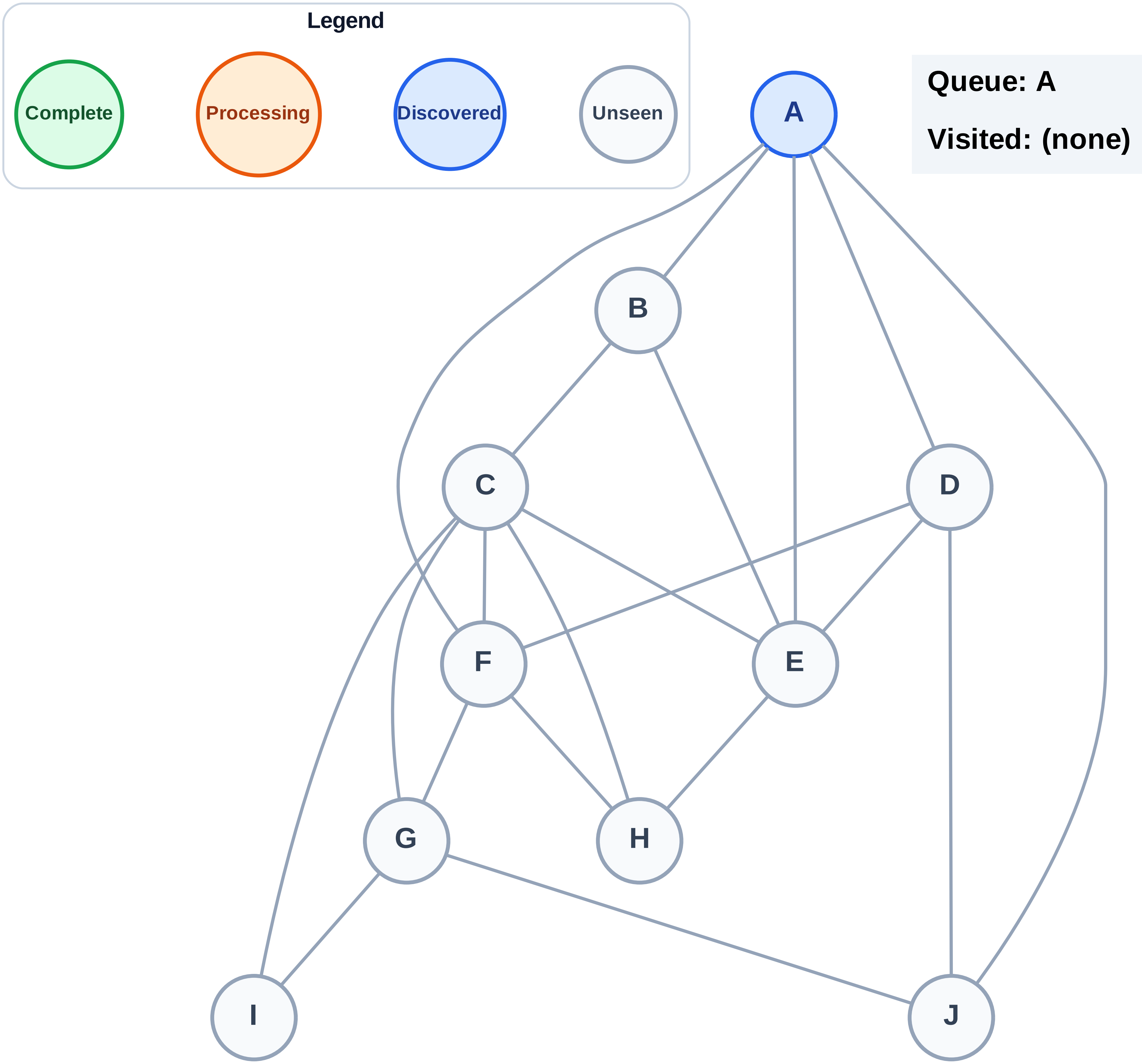
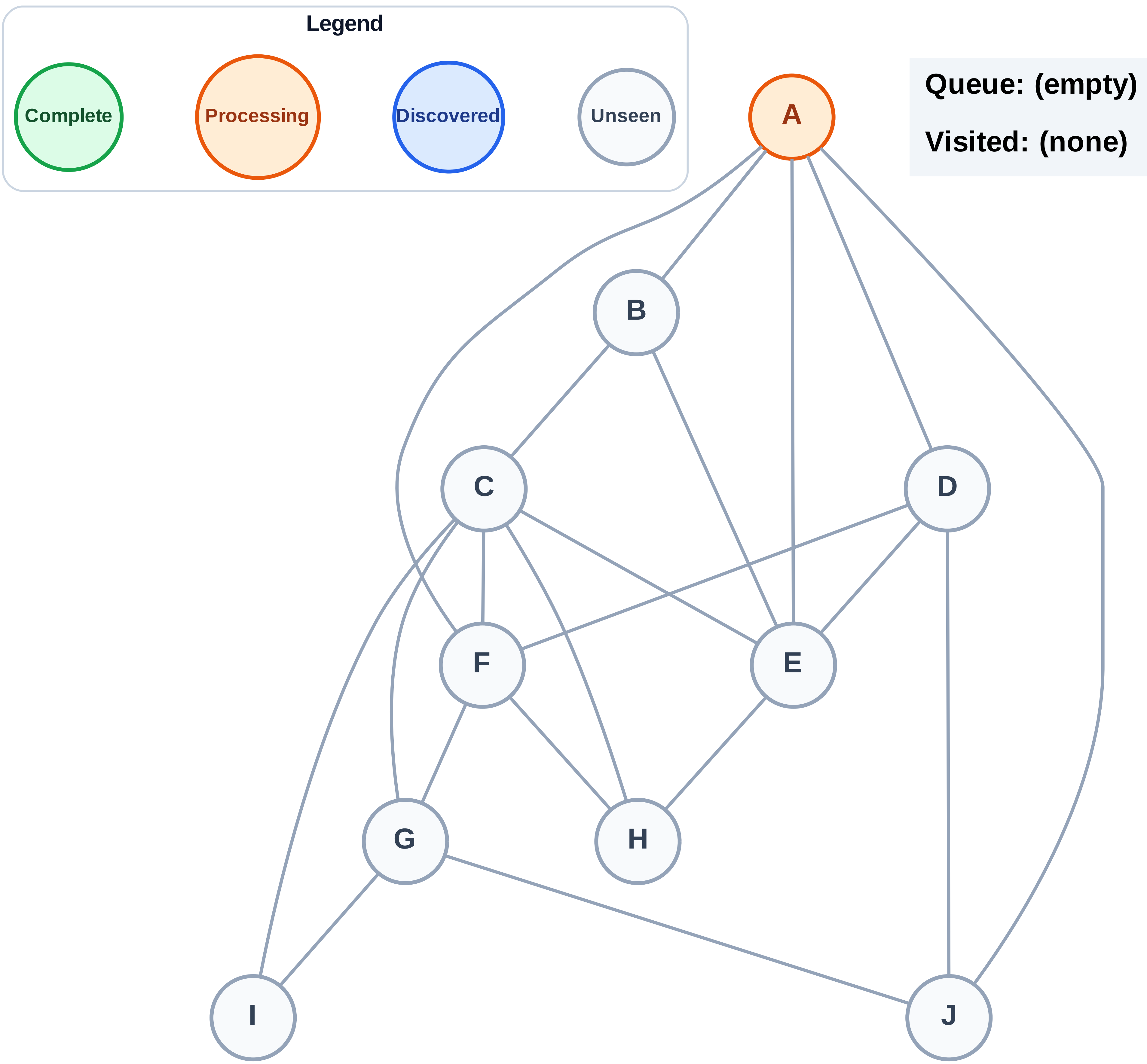
BFS Traversal

Step 1: Start at A



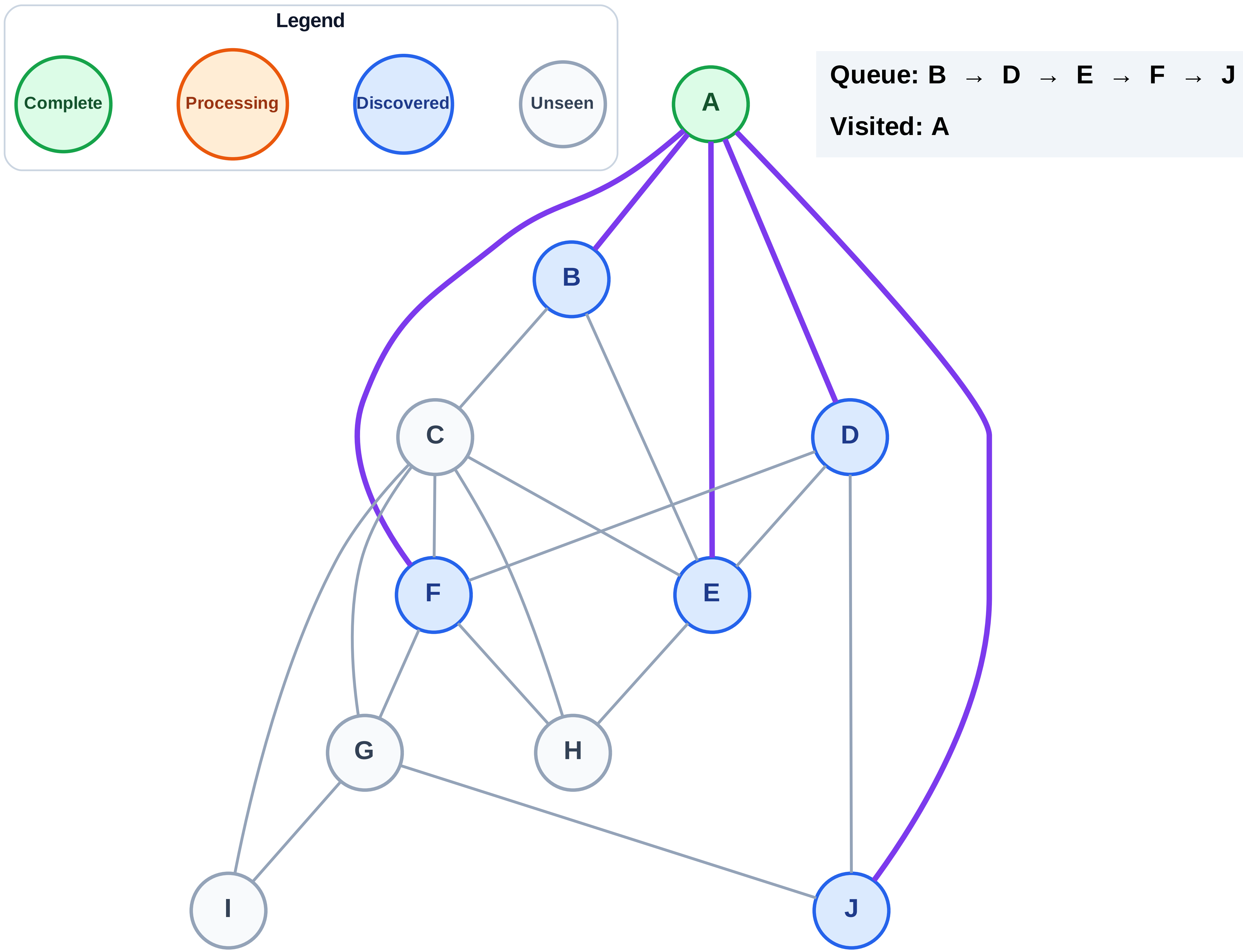
BFS Traversal

Step 2: Process node A



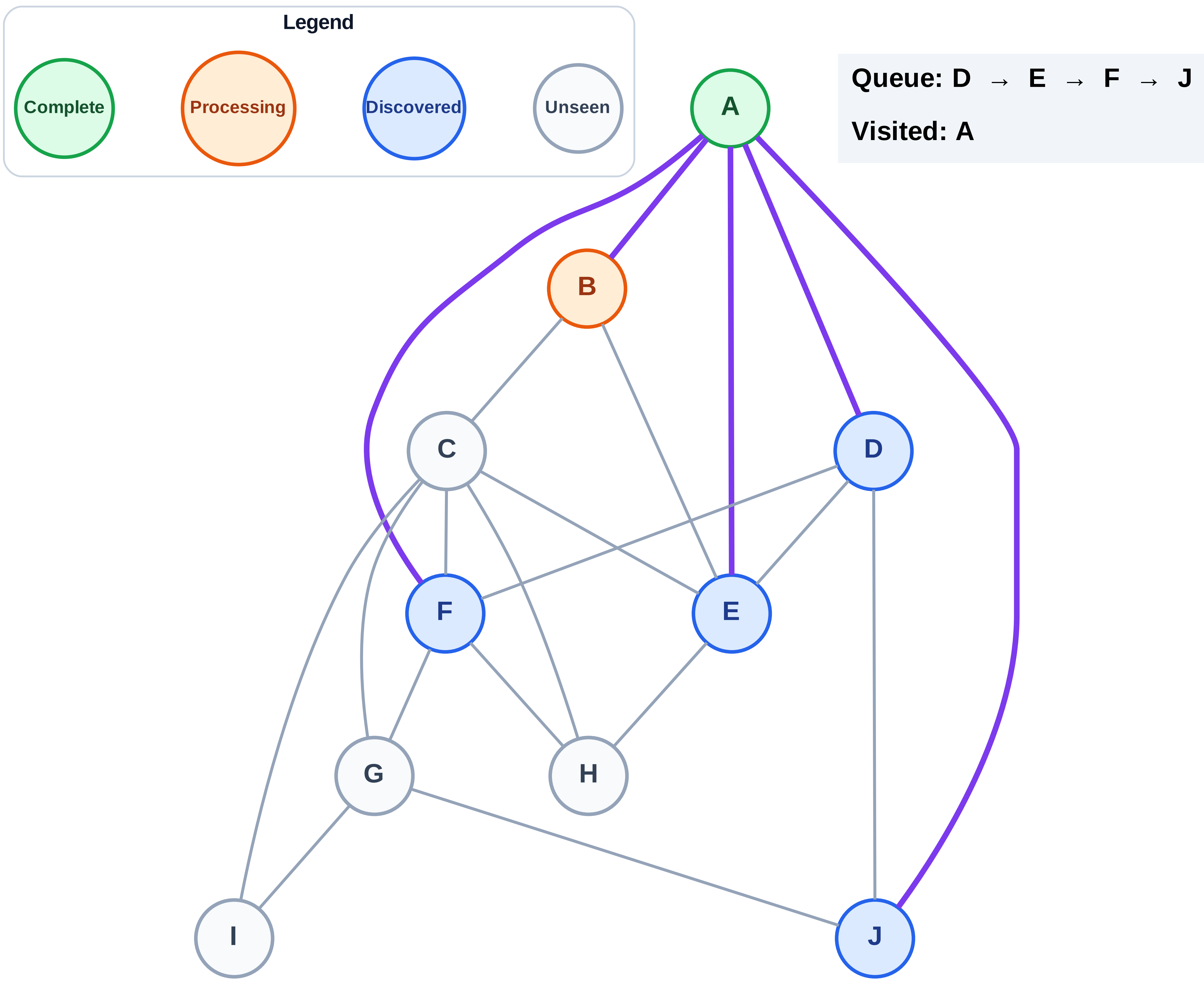
BFS Traversal

Step 3: Discover B, D, E, F, J from A



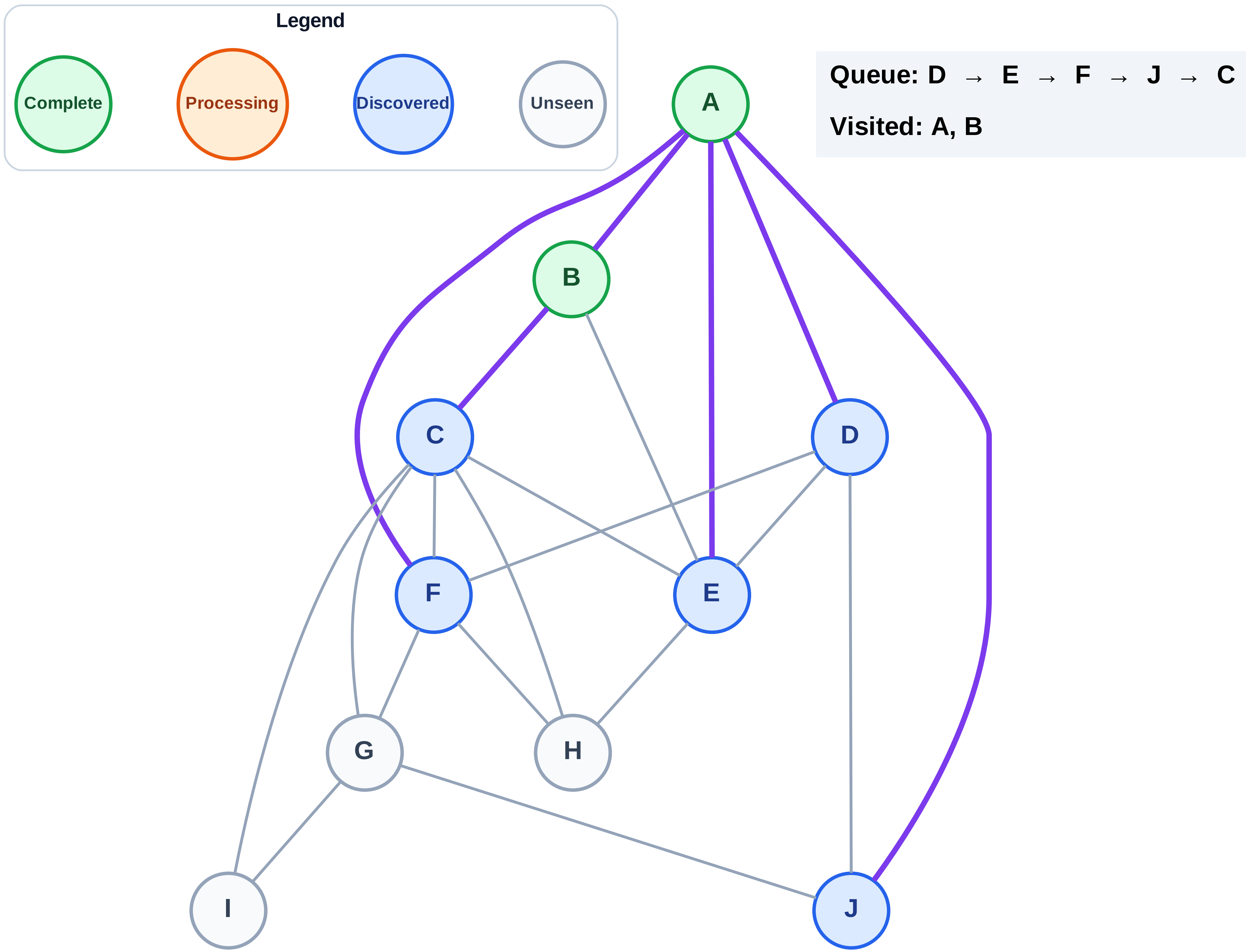
BFS Traversal

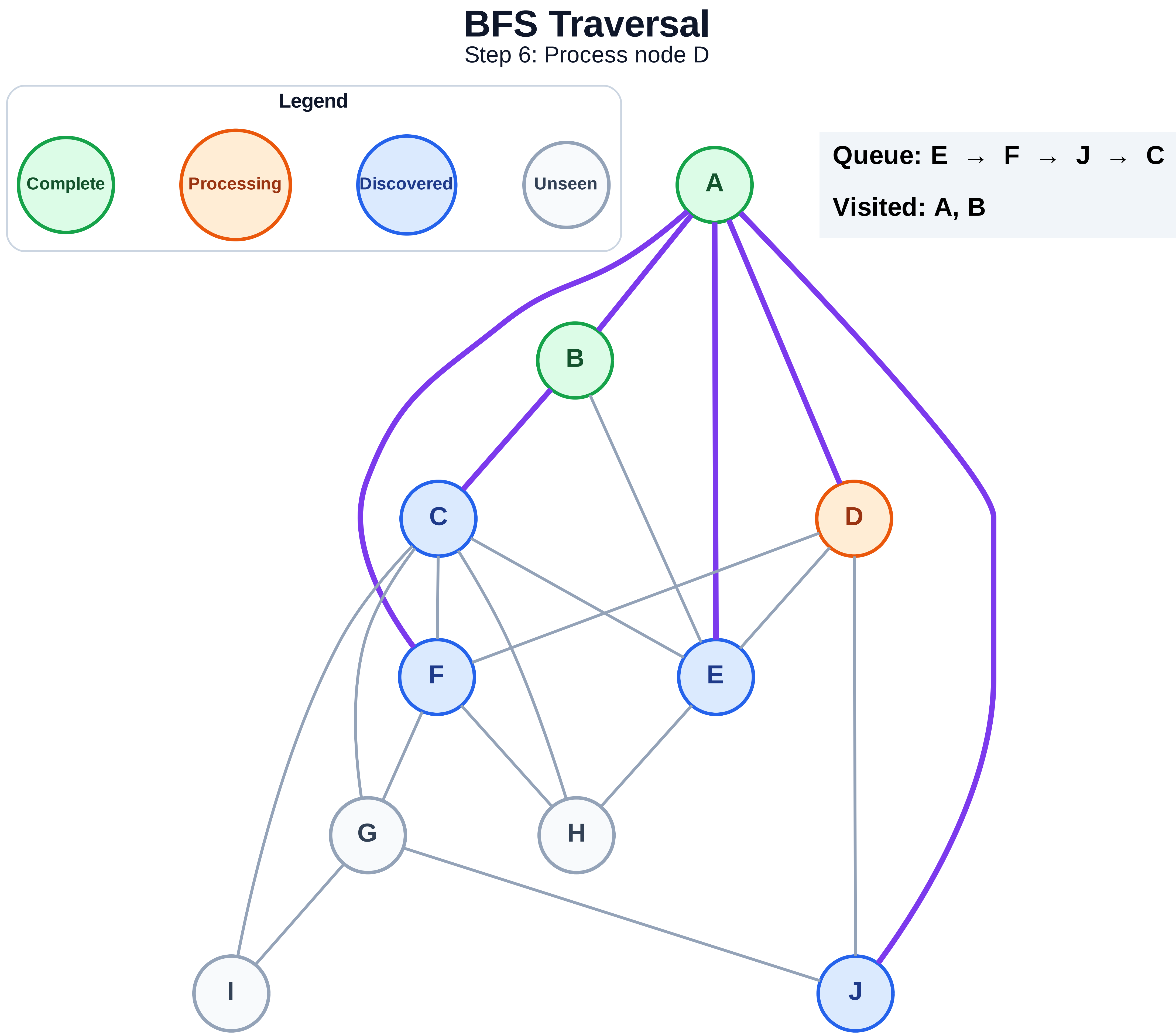
Step 4: Process node B



BFS Traversal

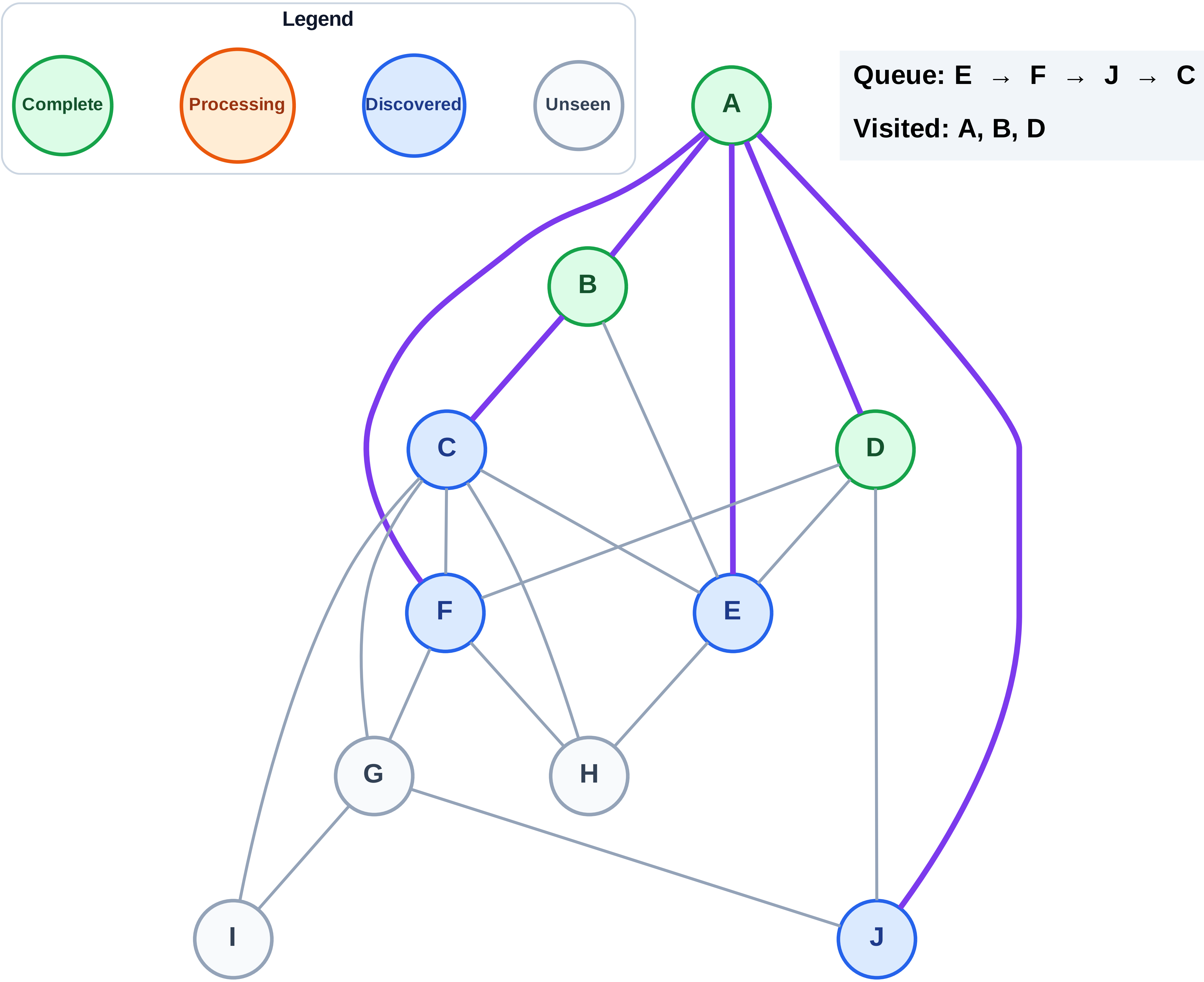
Step 5: Discover C from B




BFS Traversal
Step 6: Process node D
Legend
Complete
Processing
Discovered
Unseen
Queue: E → F → J → C
Visited: A, B
A
B
C
D
E
F
G
H
I
J

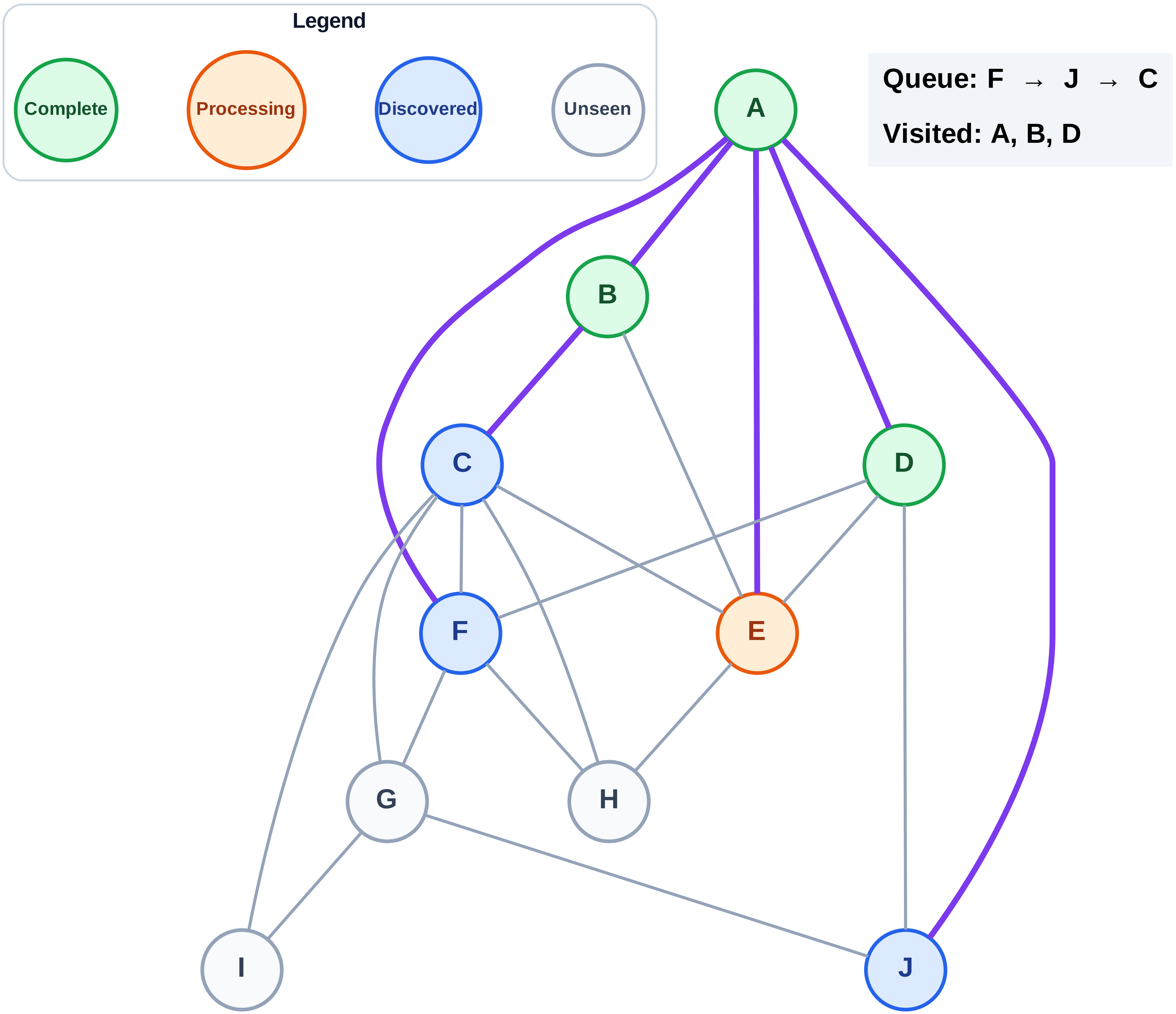
BFS Traversal

Step 7: No new neighbors from D



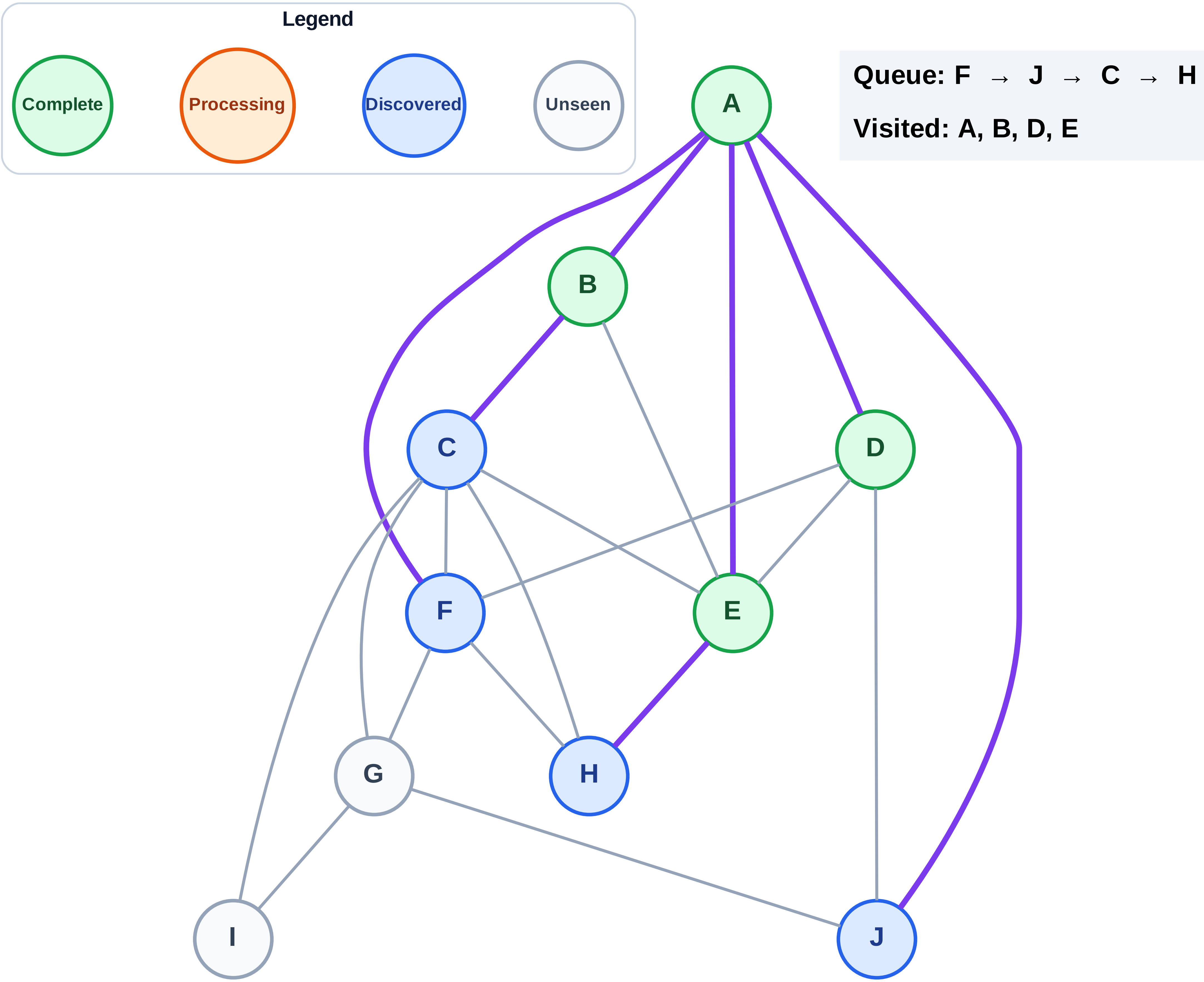
BFS Traversal

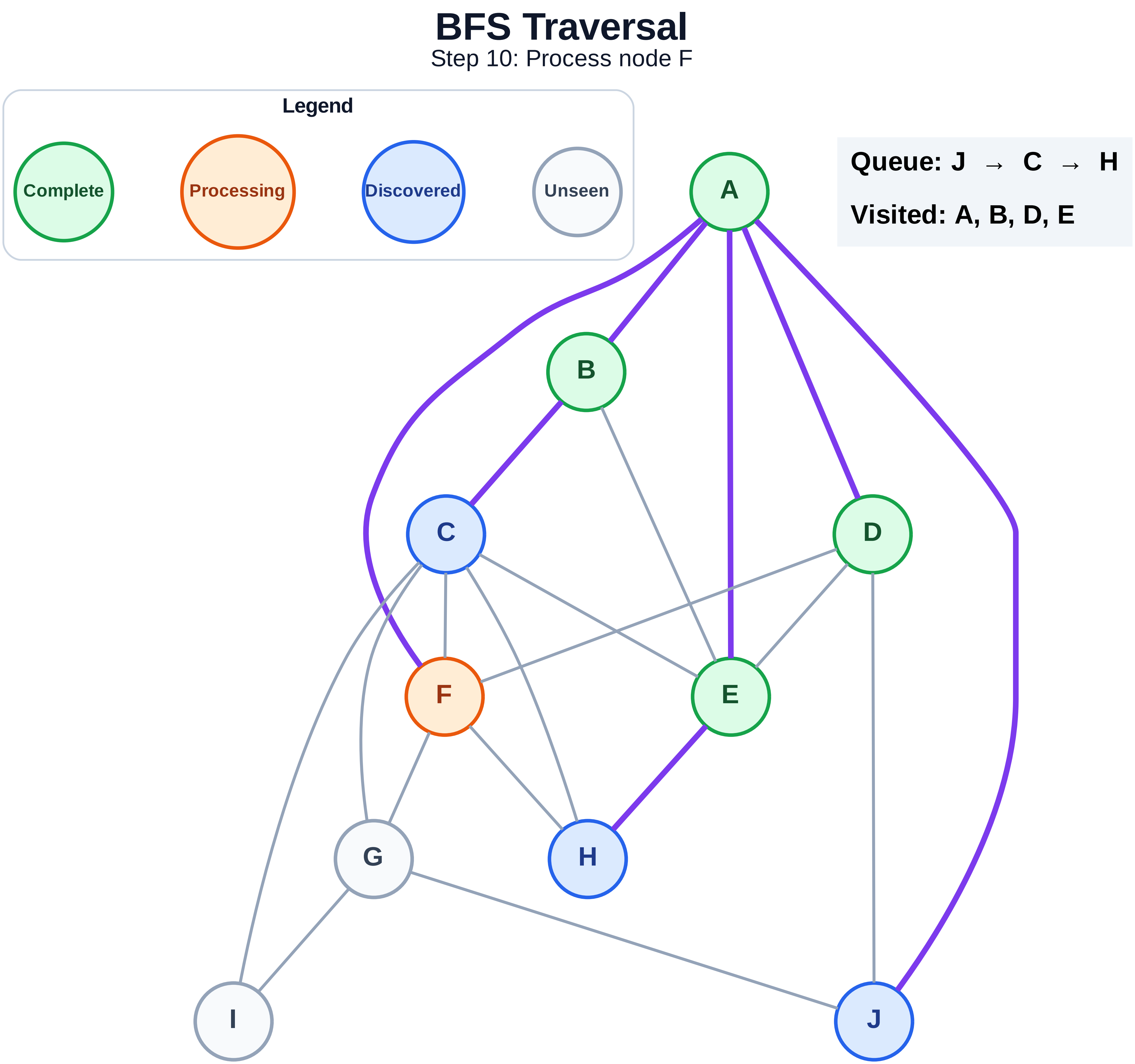
Step 8: Process node E



BFS Traversal

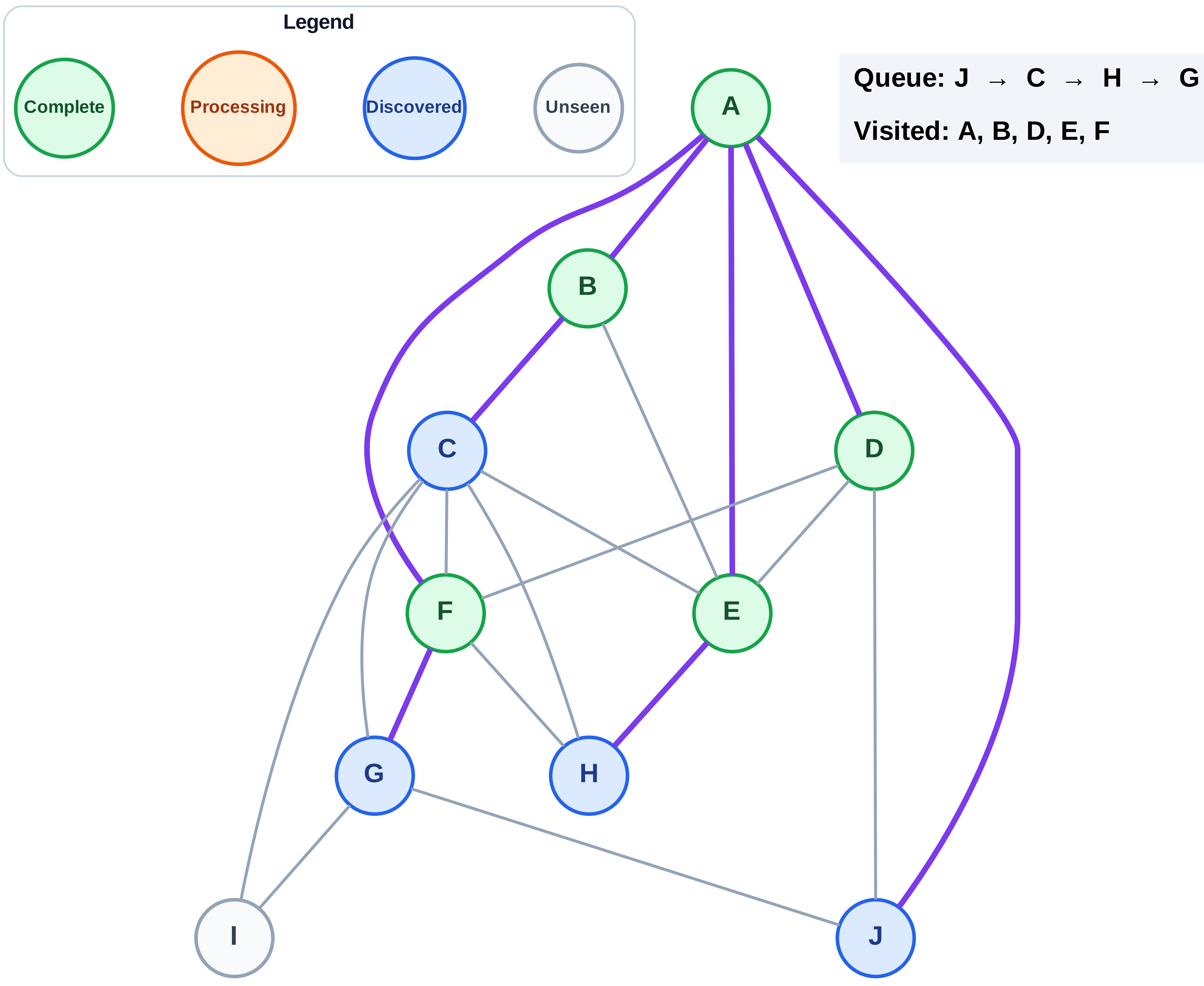
Step 9: Discover H from E




BFS Traversal
Step 10: Process node F
Legend
Complete
Processing
Discovered
Unseen
Queue: J → C → H
Visited: A, B, D, E
A
B
C
D
E
F
G
H
I
J

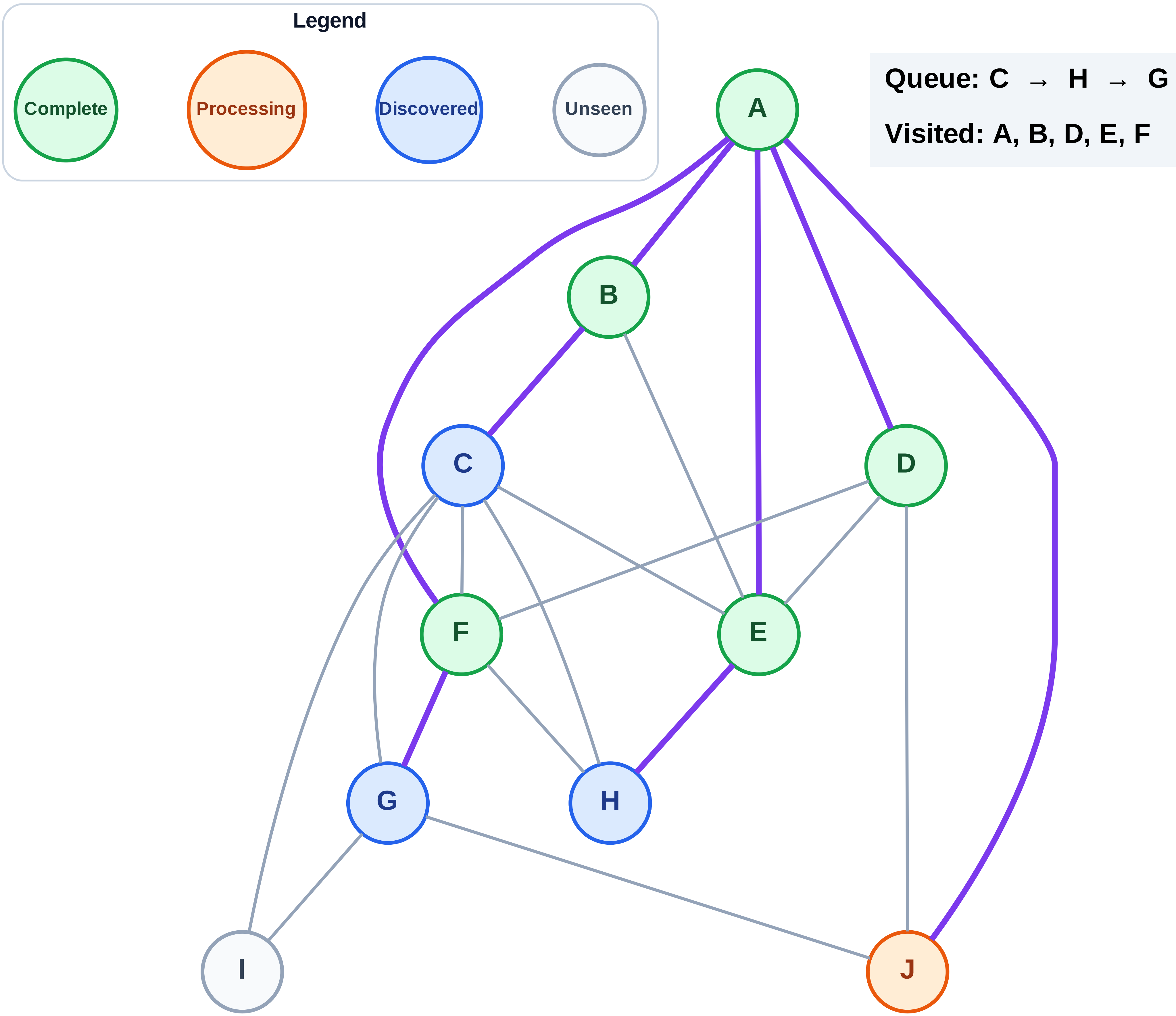
BFS Traversal

Step 11: Discover G from F



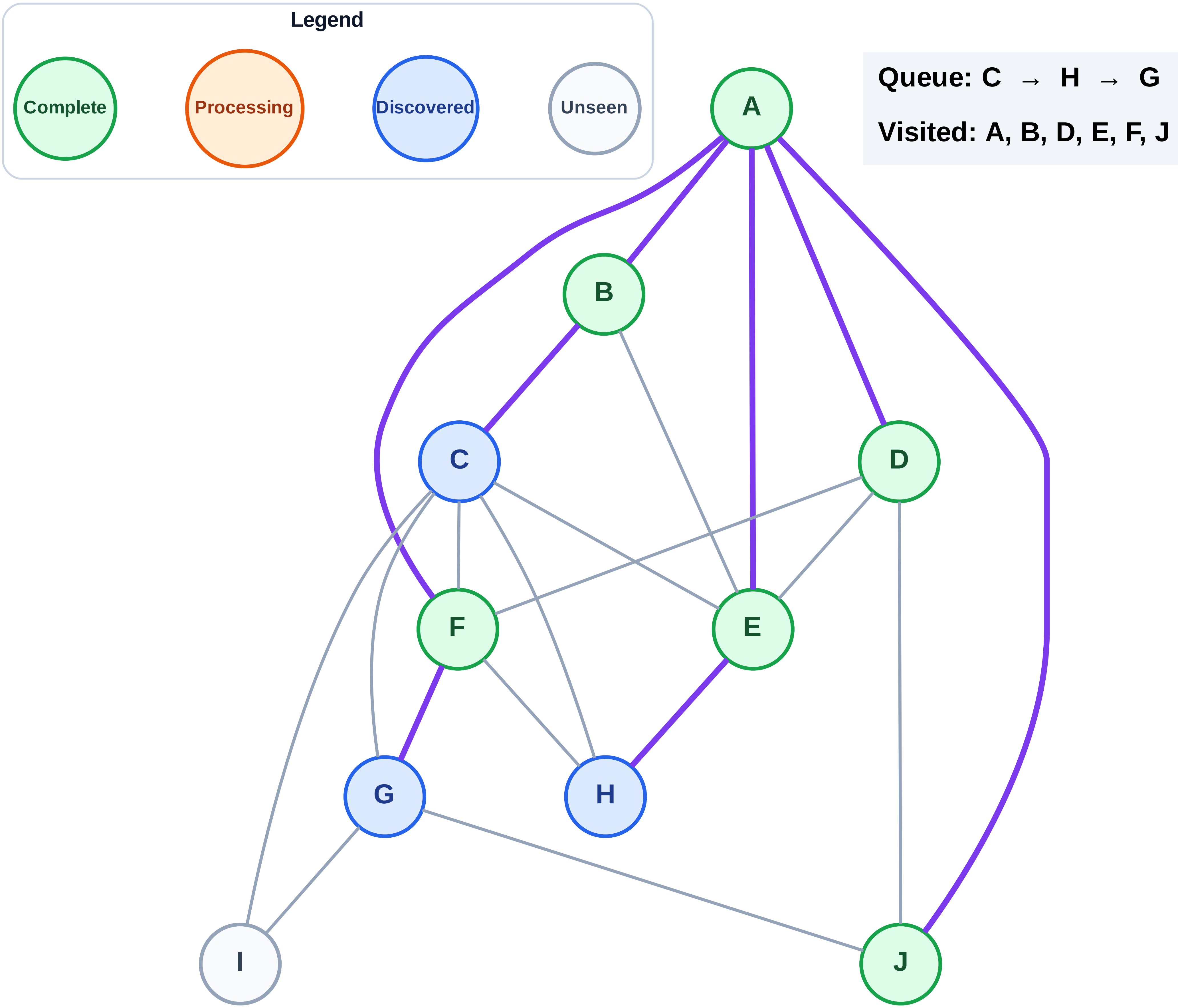
BFS Traversal

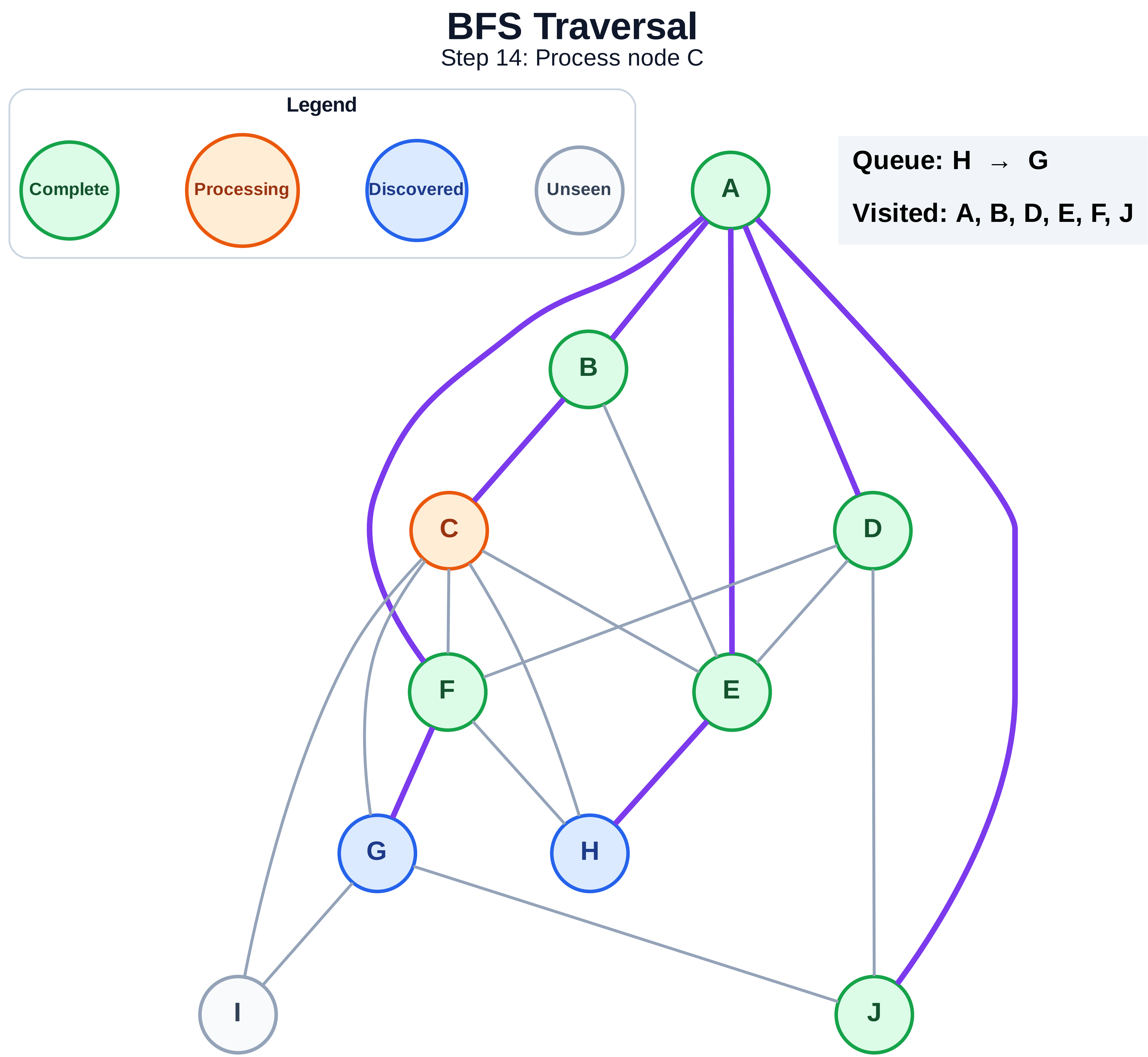
Step 12: Process node J



BFS Traversal

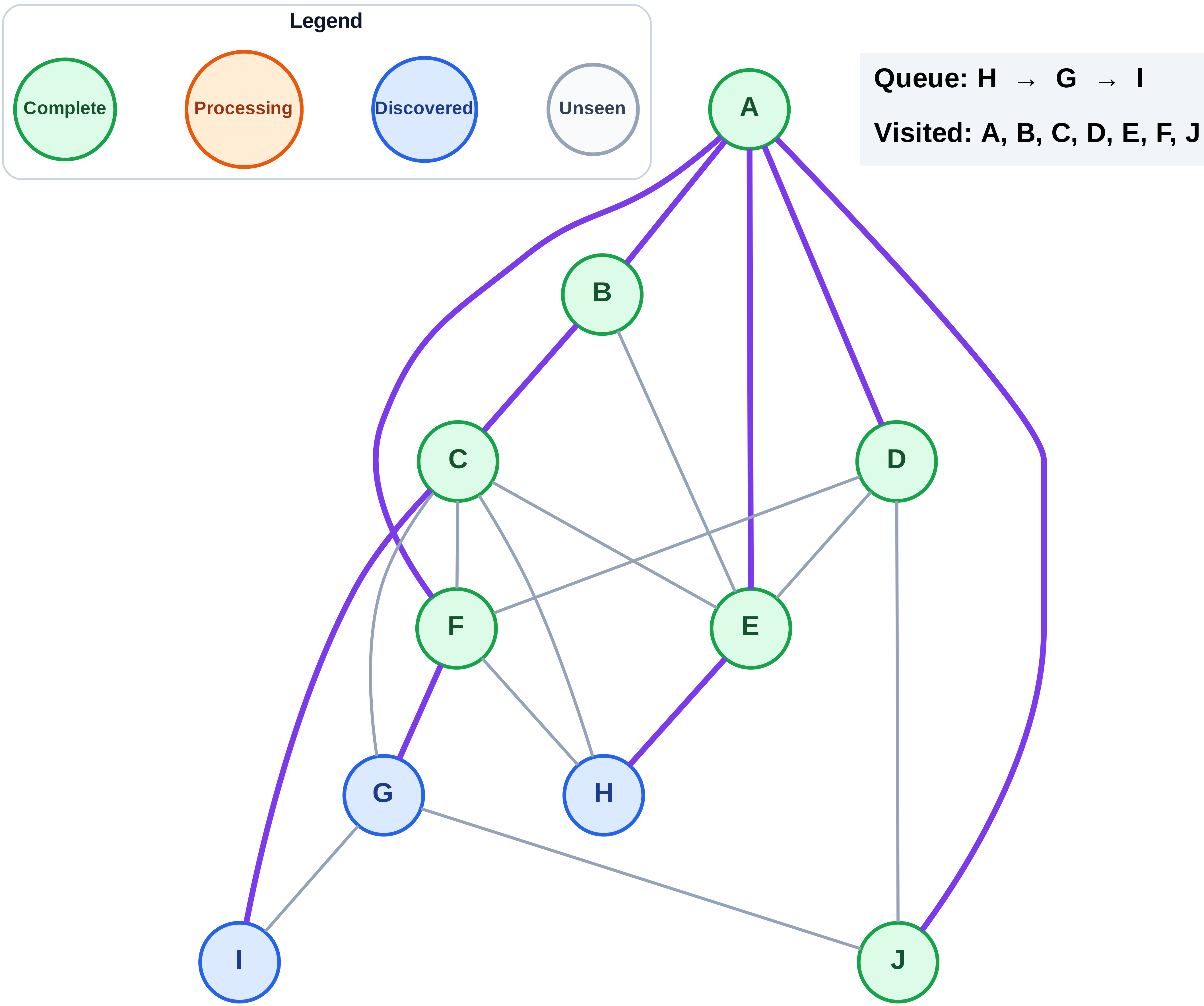
Step 13: No new neighbors from J




BFS Traversal
Step 14: Process node C
Legend
Complete
Processing
Discovered
Unseen
Queue: H → G
Visited: A, B, D, E, F, J
A
B
C
D
E
F
G
H
I
J

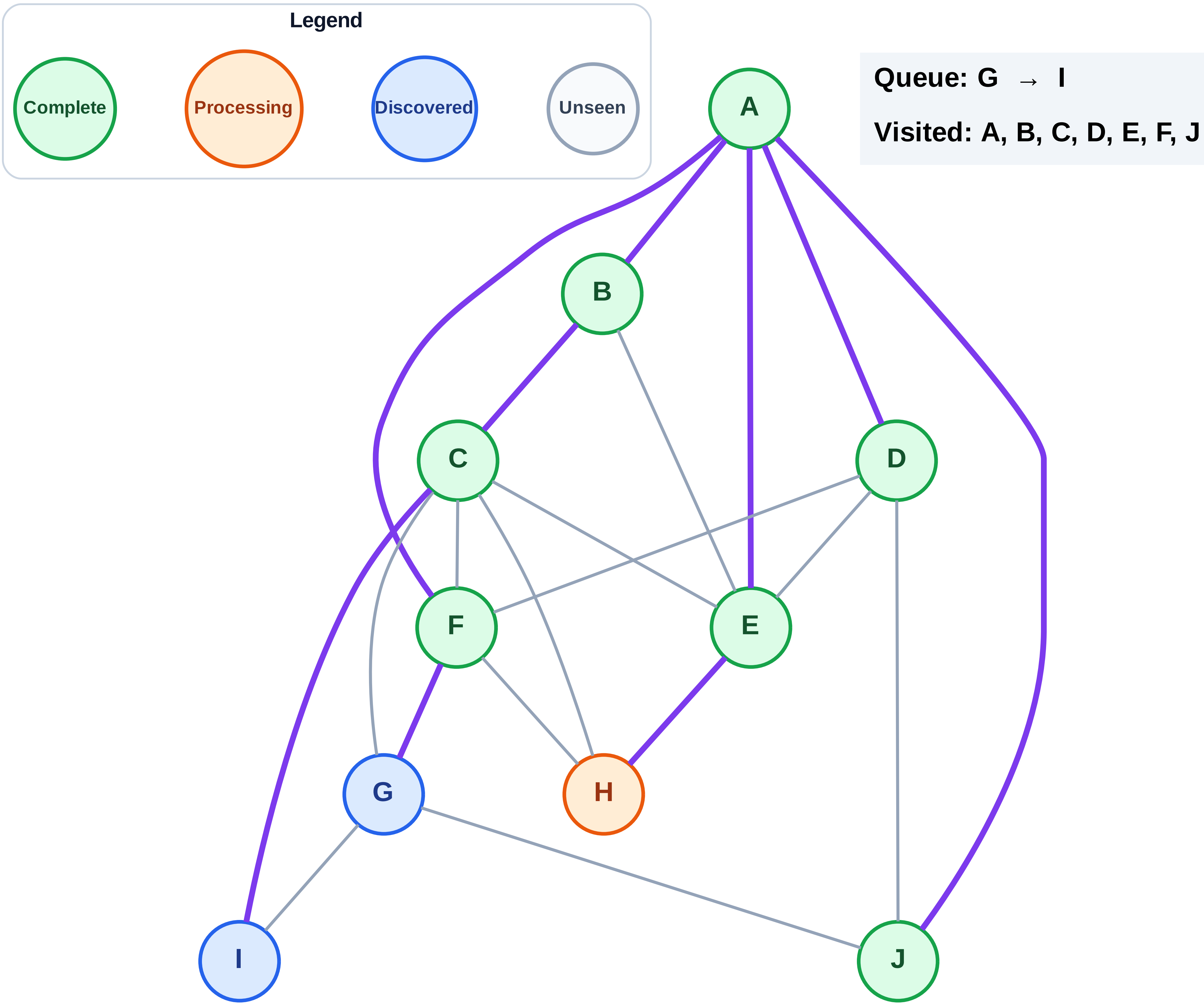
BFS Traversal

Step 15: Discover I from C



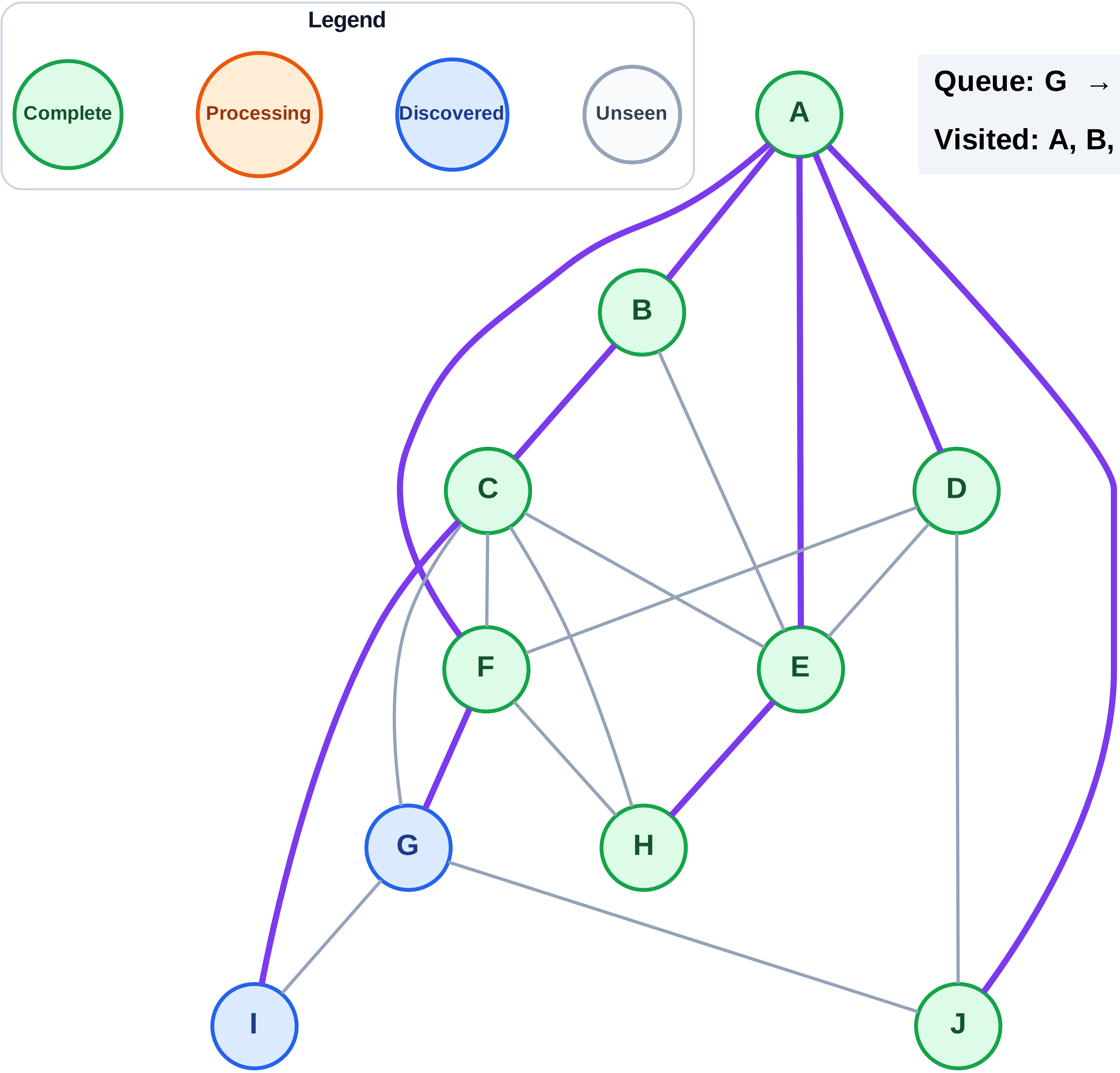
BFS Traversal

Step 16: Process node H



BFS Traversal

Step 17: No new neighbors from H

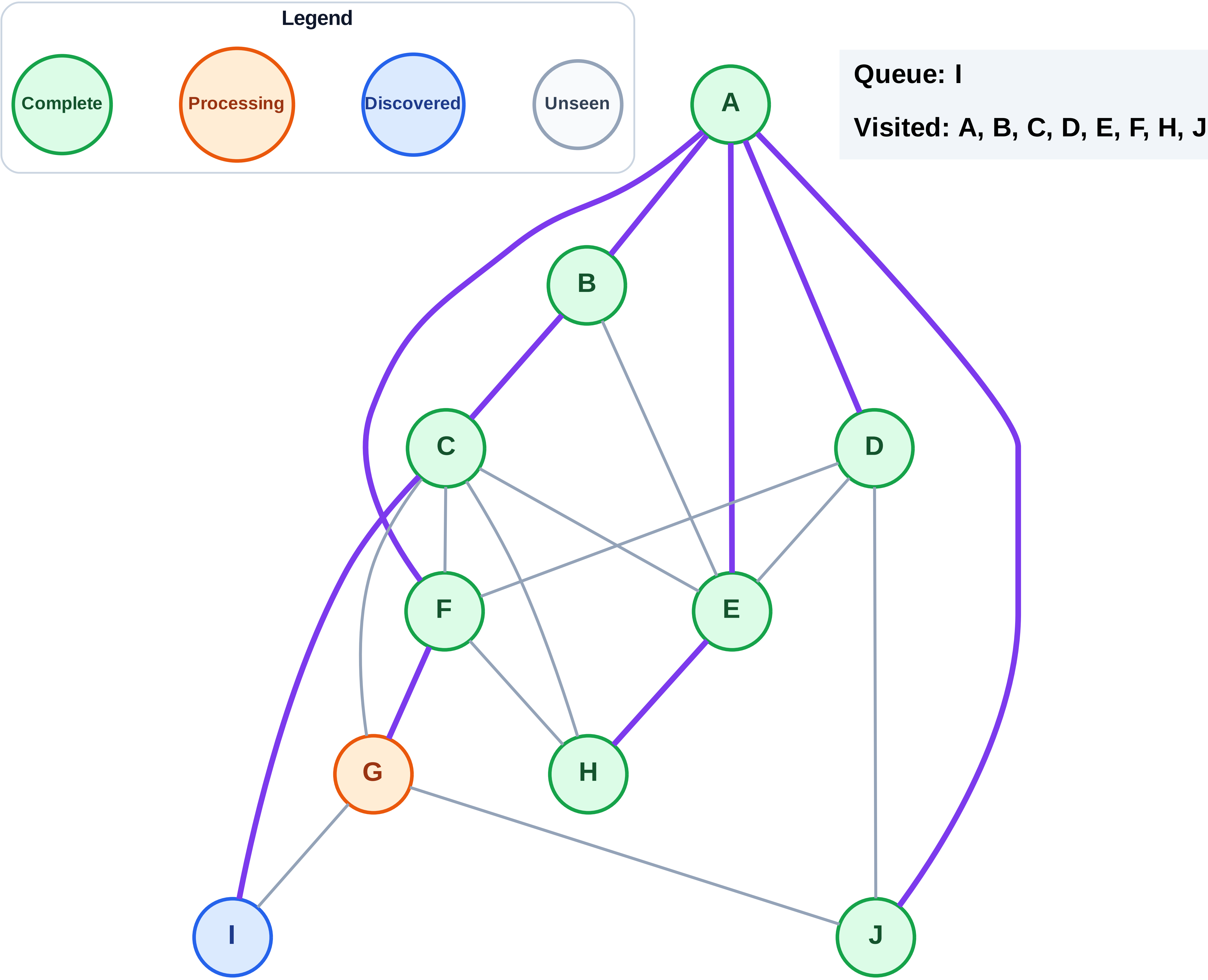


Queue: G → I

Visited: A, B, C, D, E, F, H, J

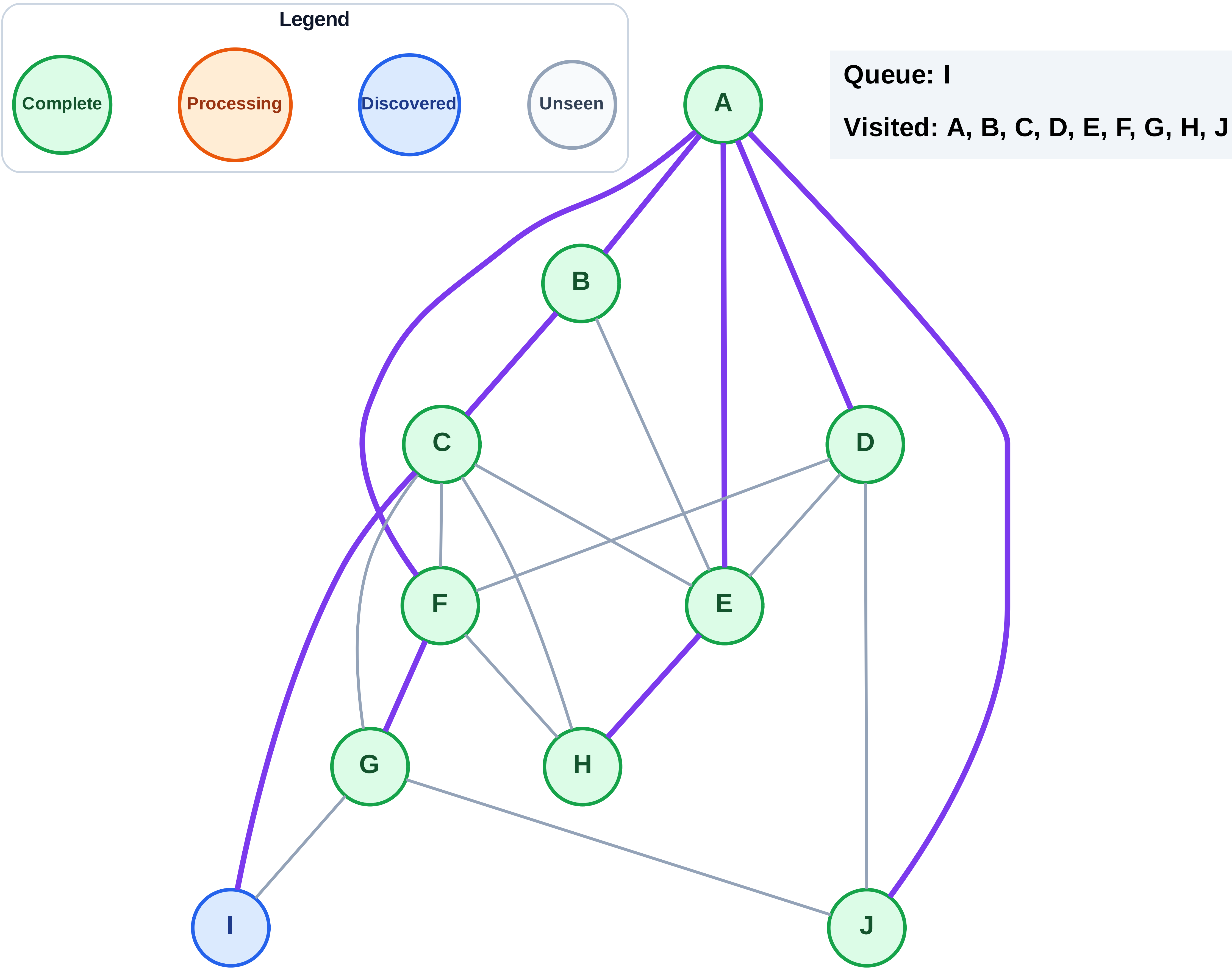
BFS Traversal

Step 18: Process node G



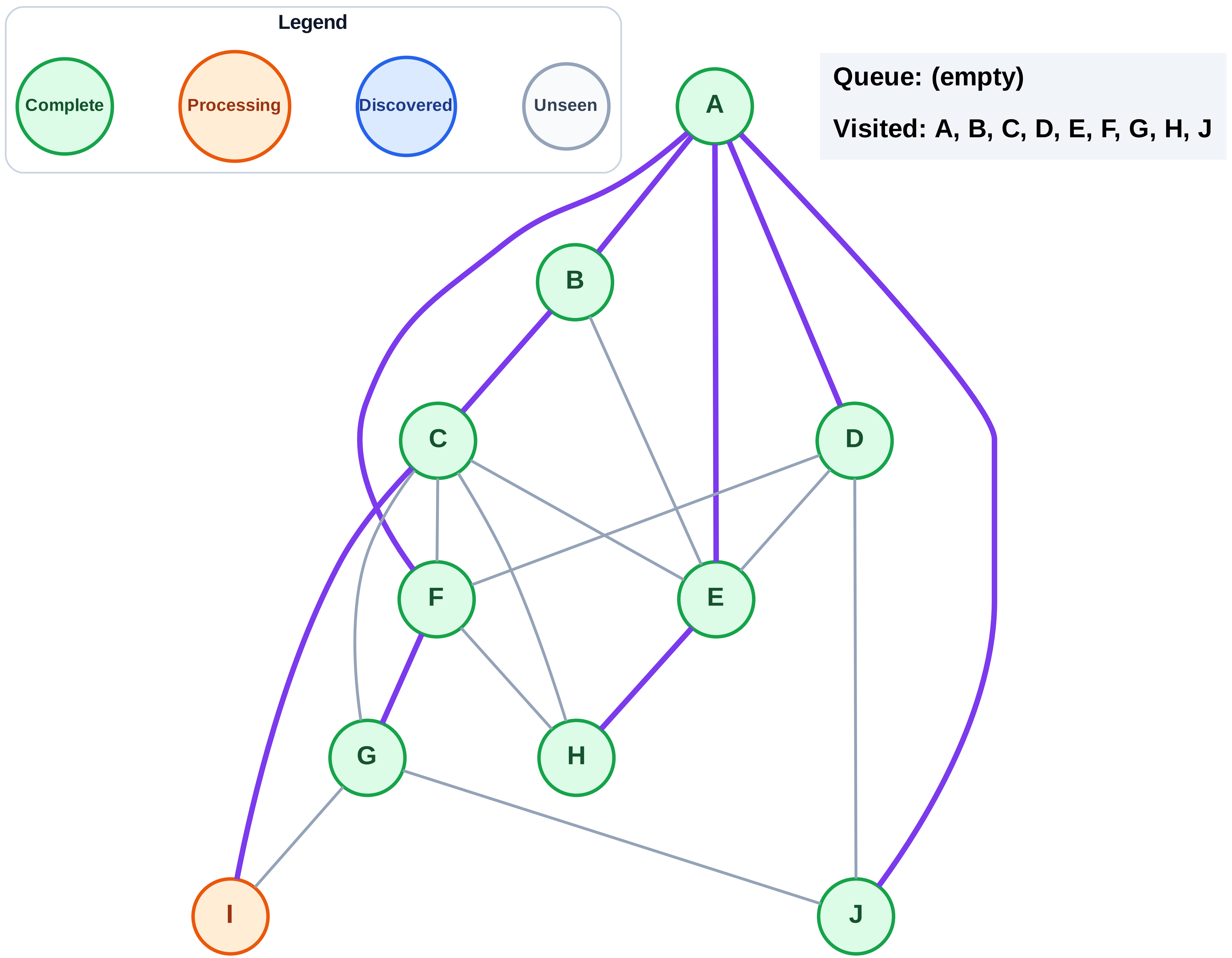
BFS Traversal

Step 19: No new neighbors from G



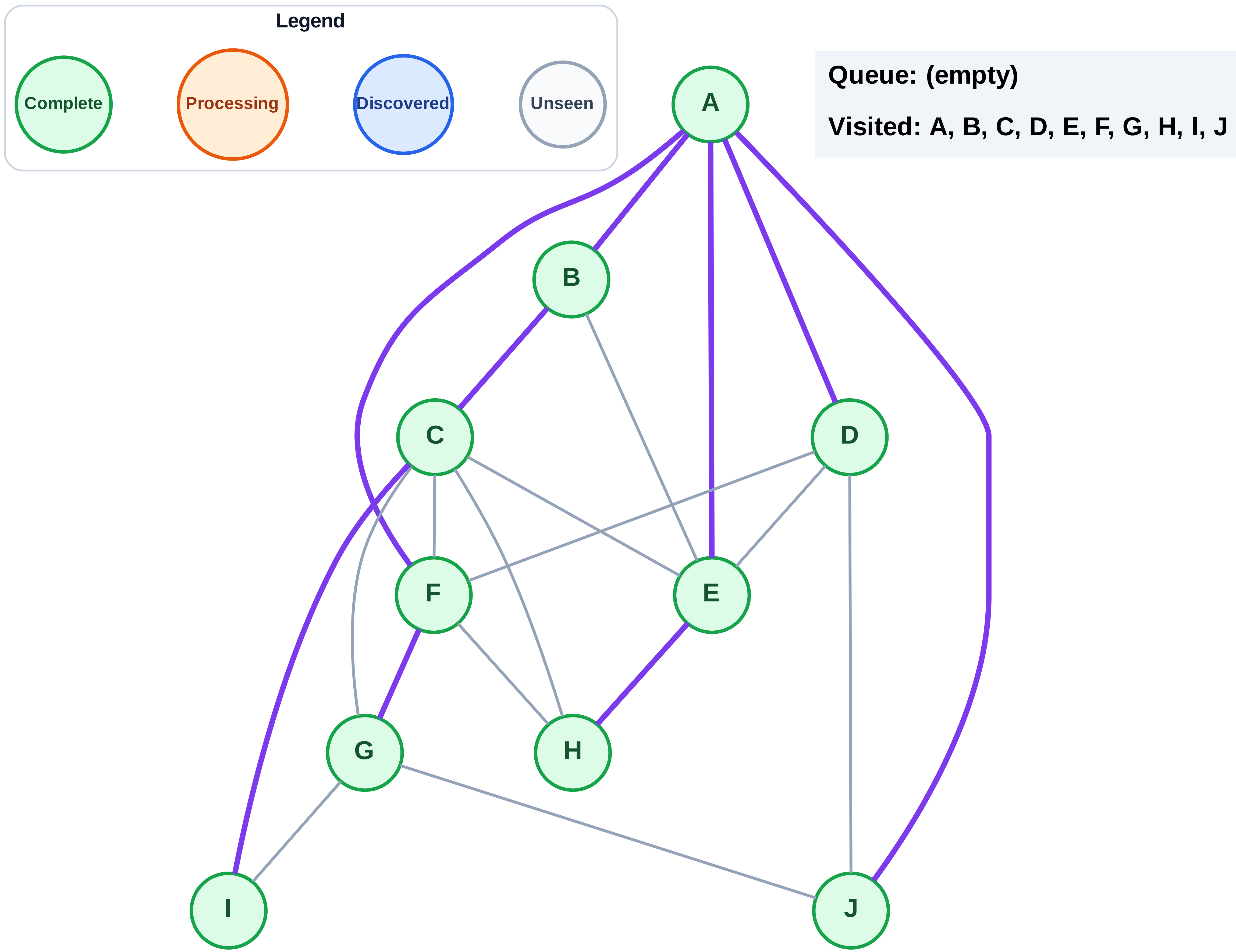
BFS Traversal

Step 20: Process node I



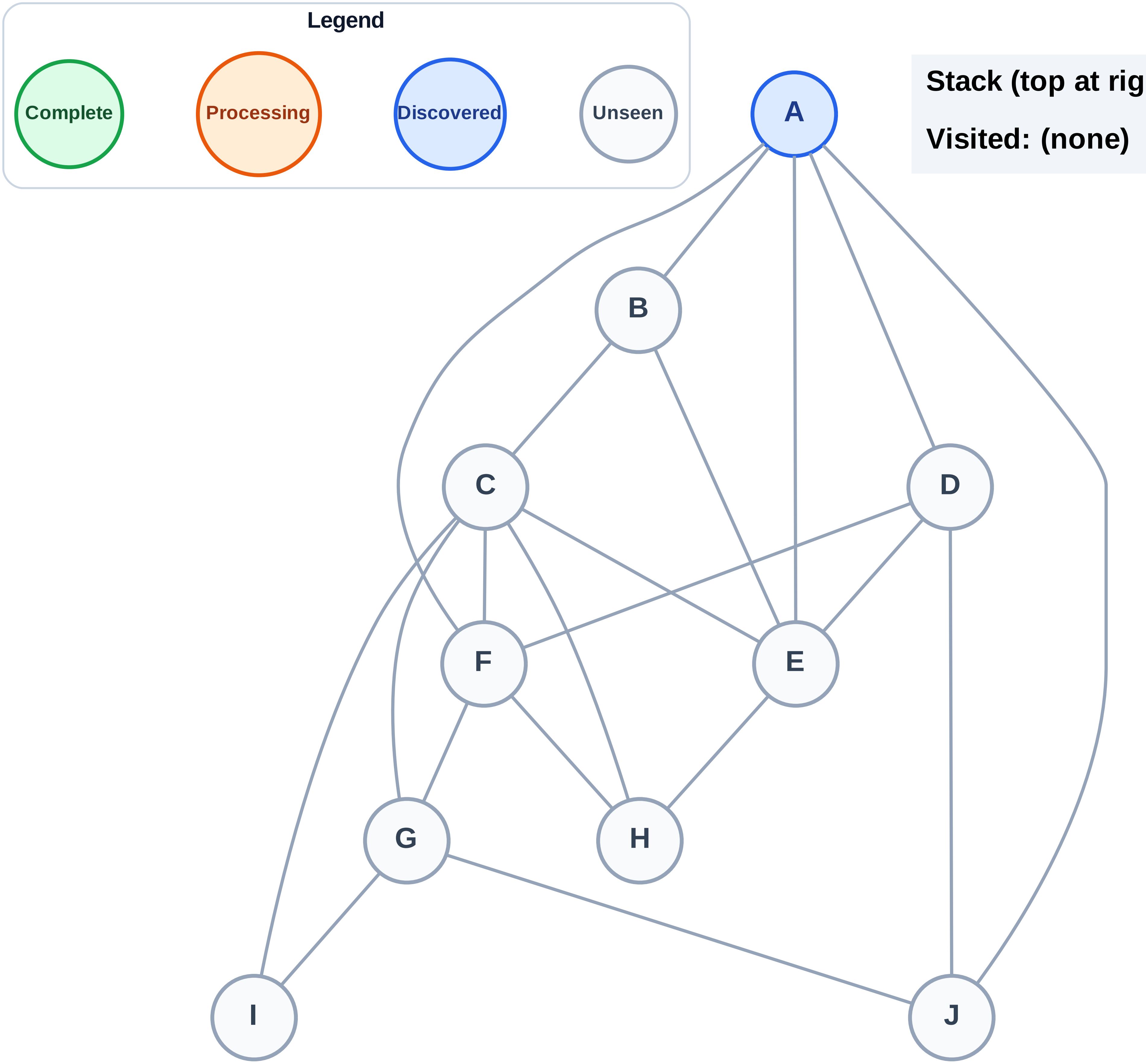
BFS Traversal

Step 21: No new neighbors from I



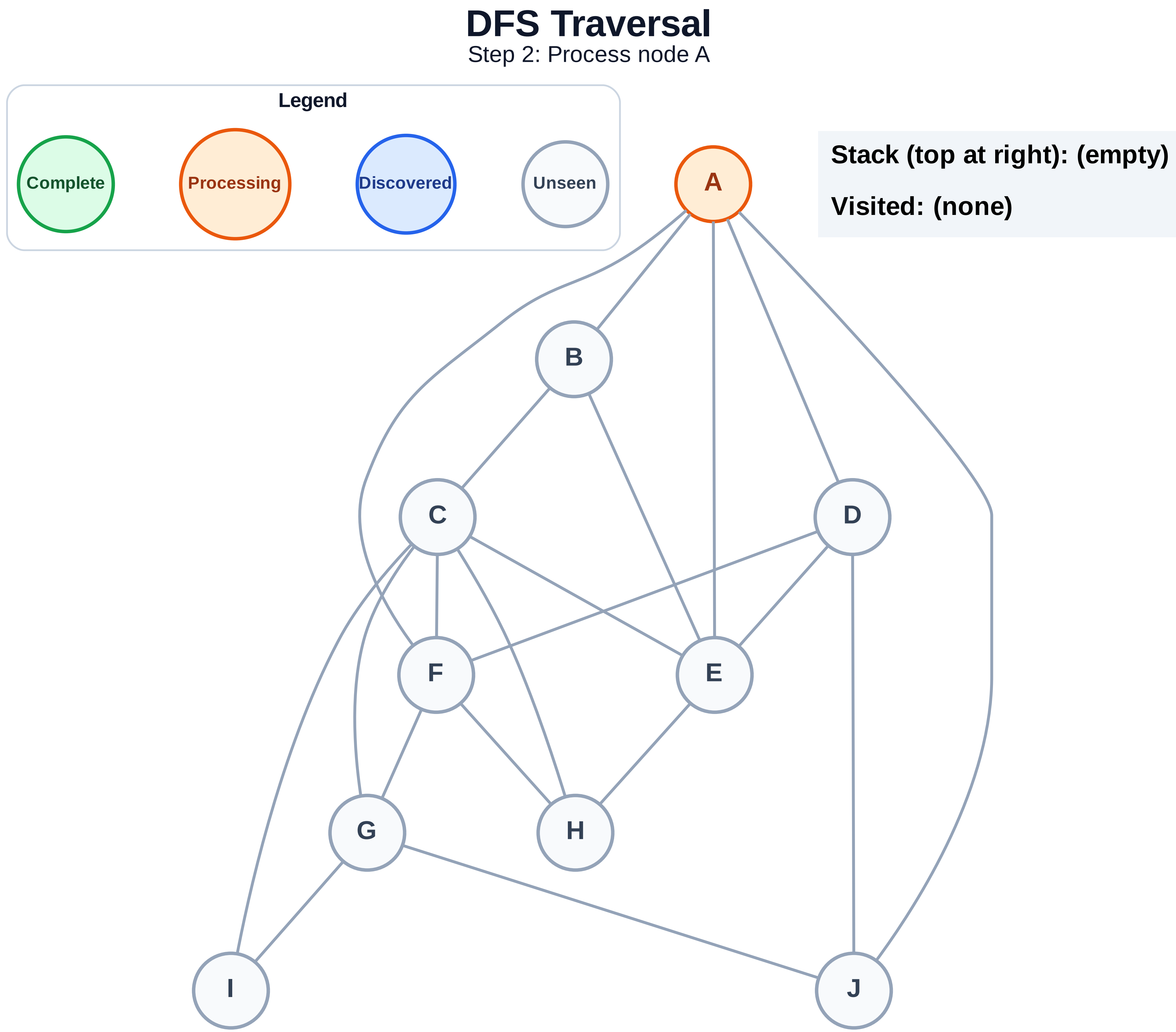
DFS Traversal

Step 1: Start at A



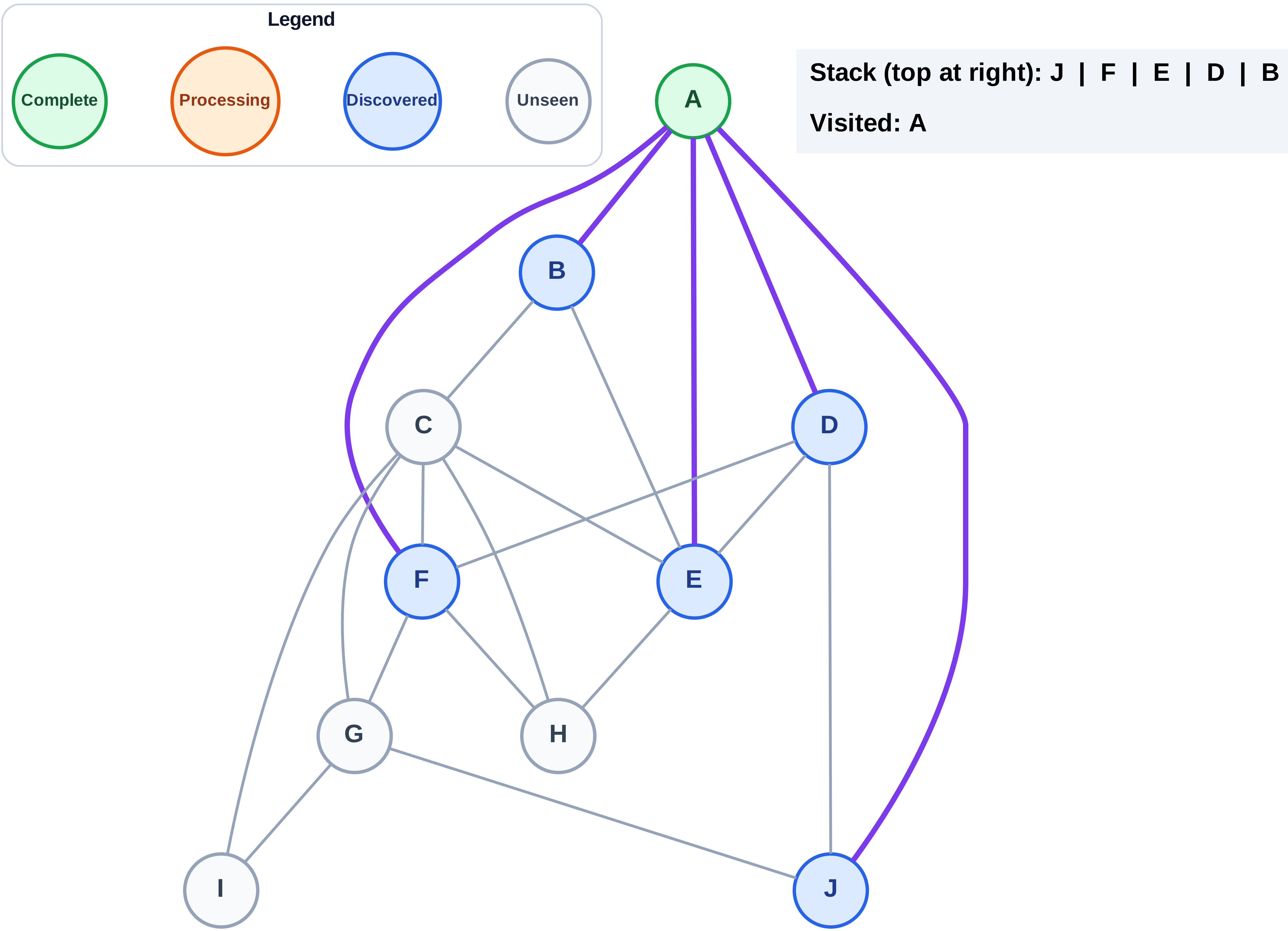
Stack (top at right): A

Visited: (none)


DFS Traversal
Step 2: Process node A
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): (empty)
Visited: (none)
A
B
C
D
E
F
G
H
I
J

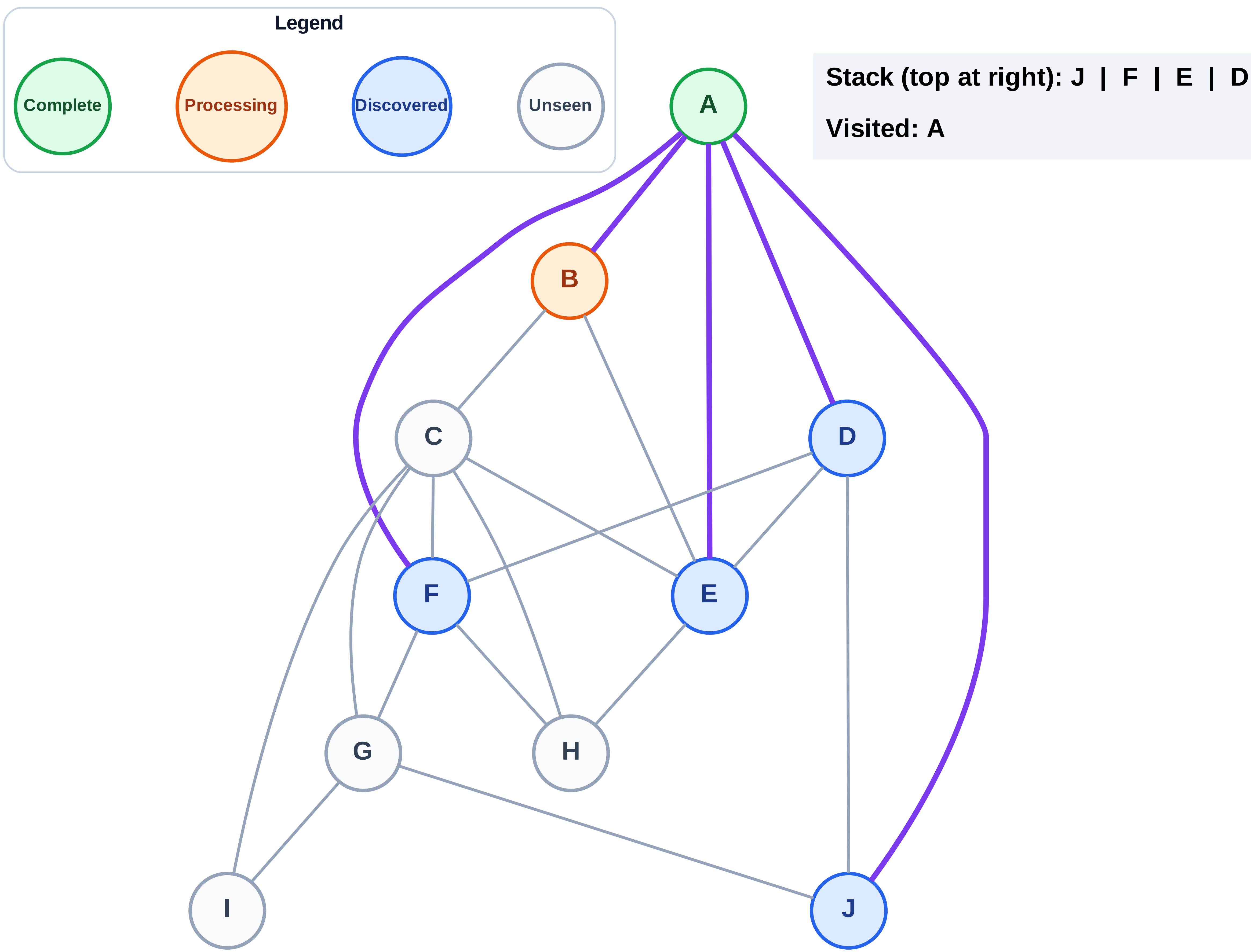
DFS Traversal

Step 3: Discover J, F, E, D, B from A



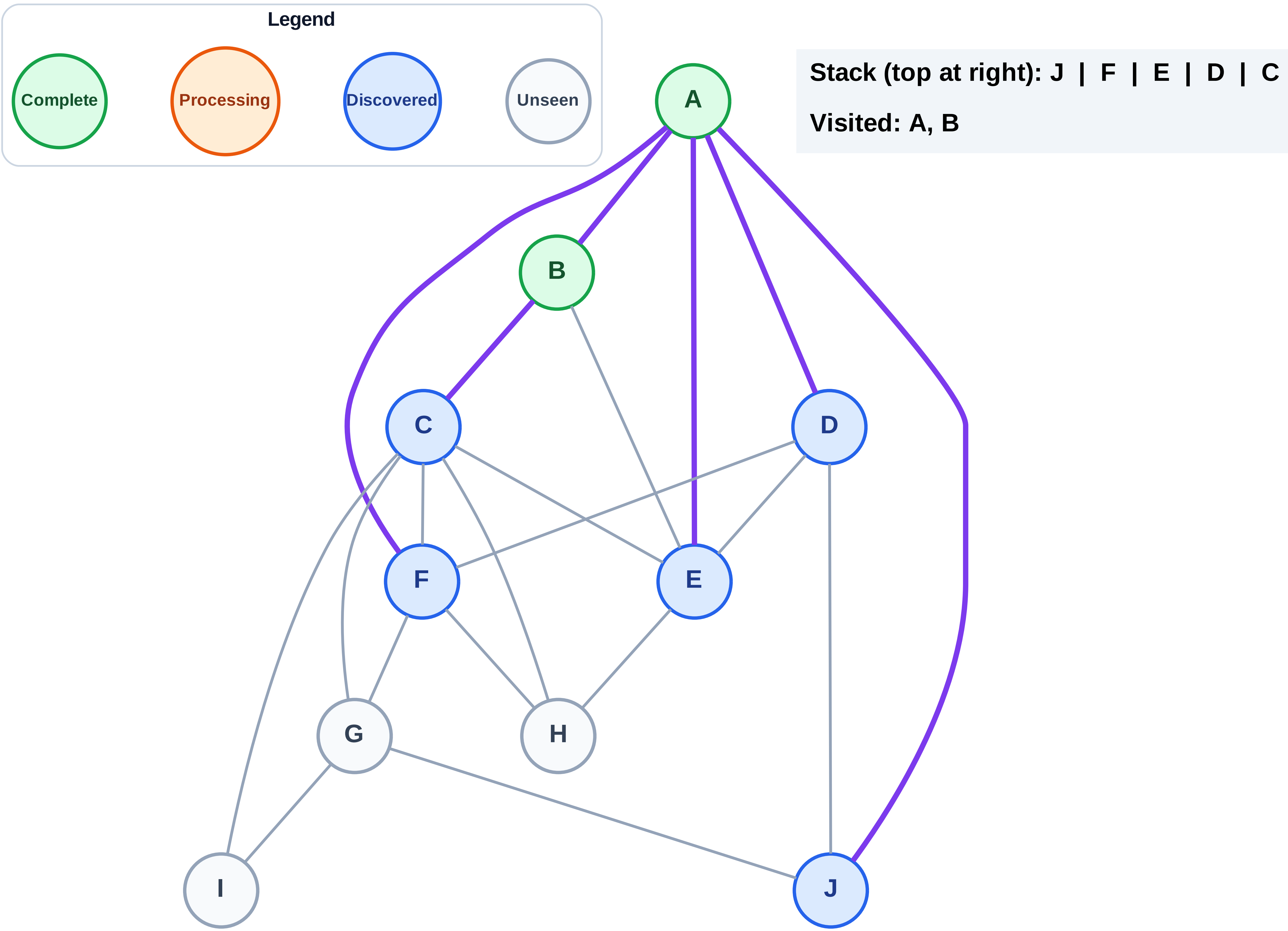
DFS Traversal

Step 4: Process node B



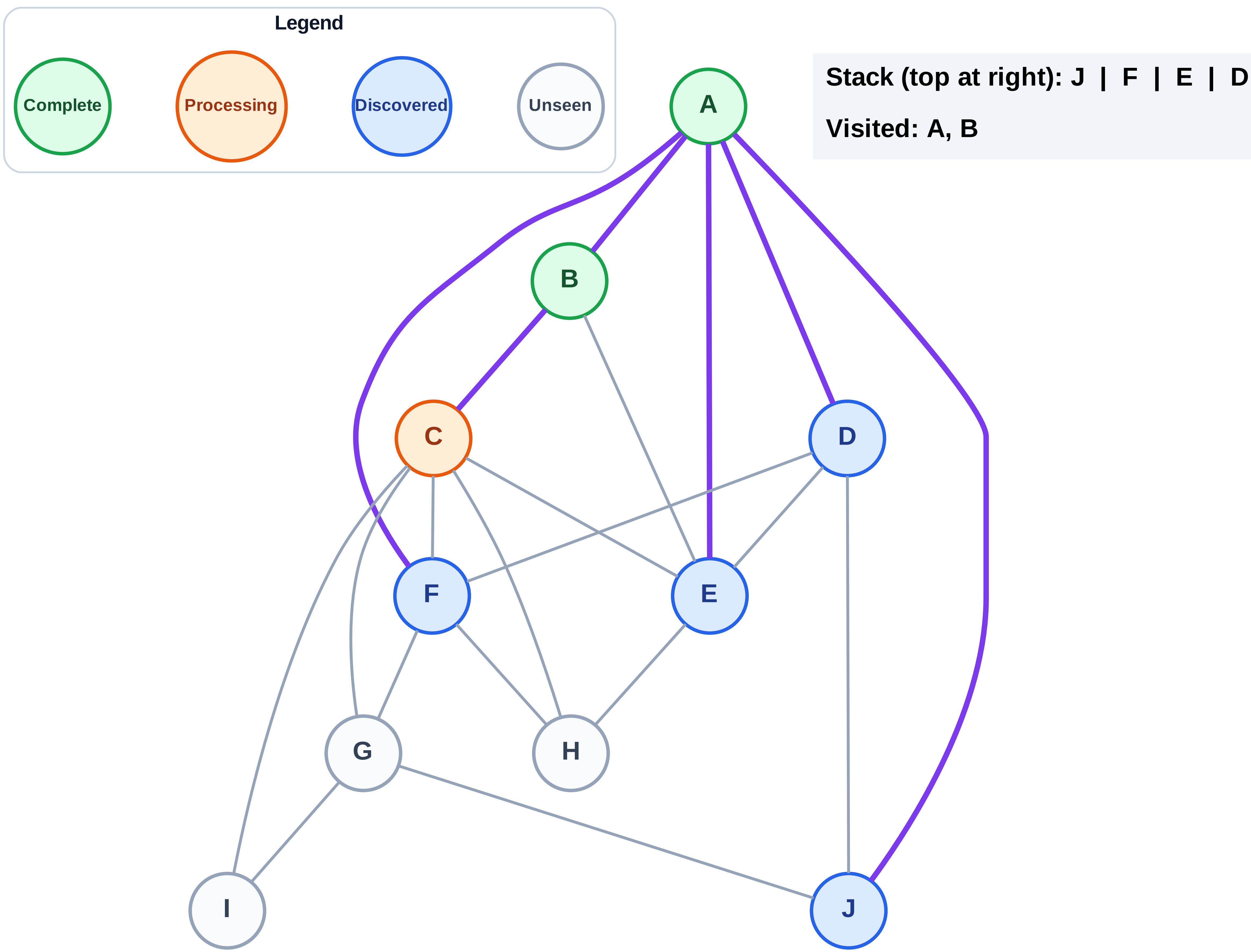
DFS Traversal

Step 5: Discover C from B



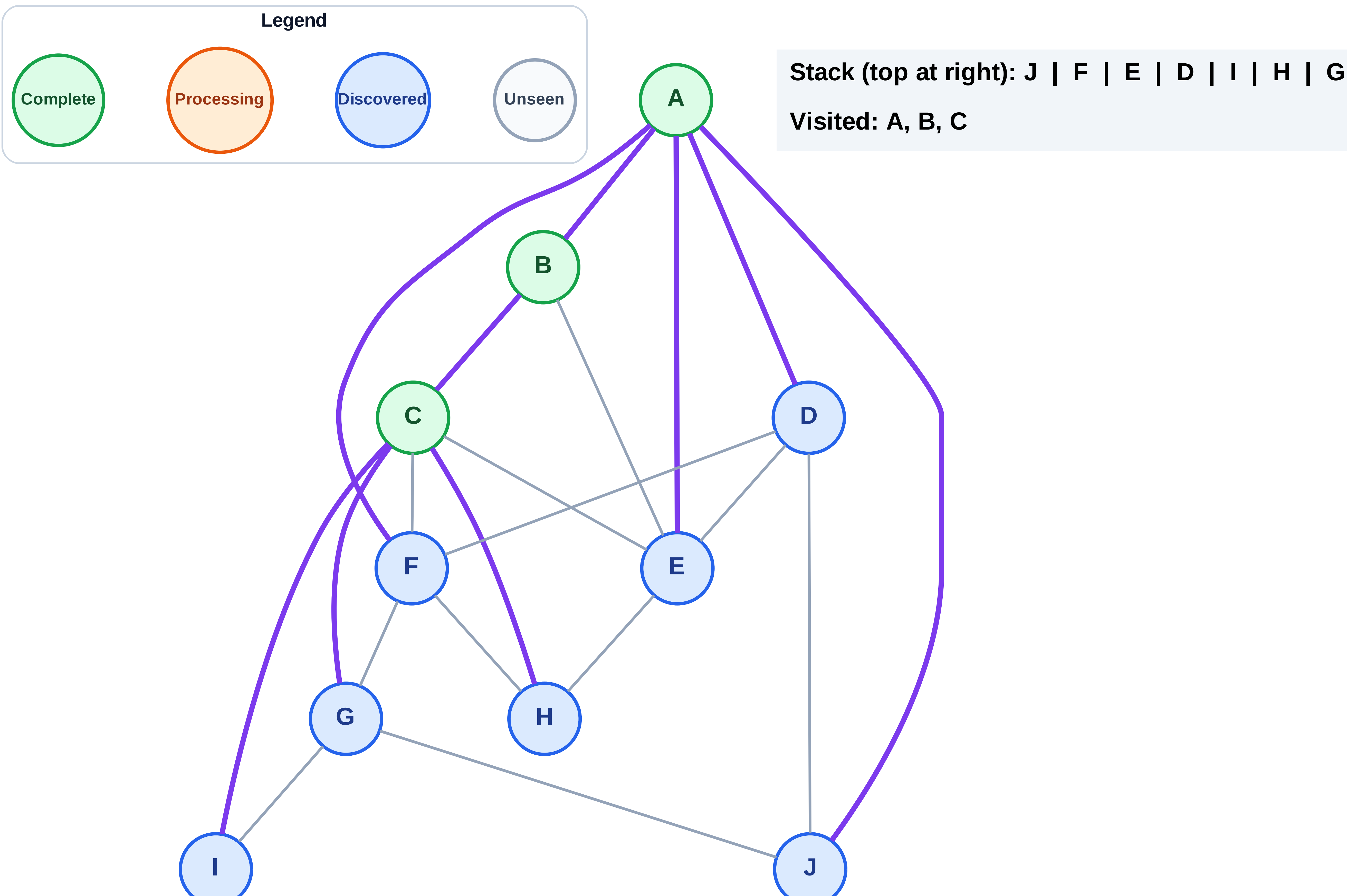
DFS Traversal

Step 6: Process node C



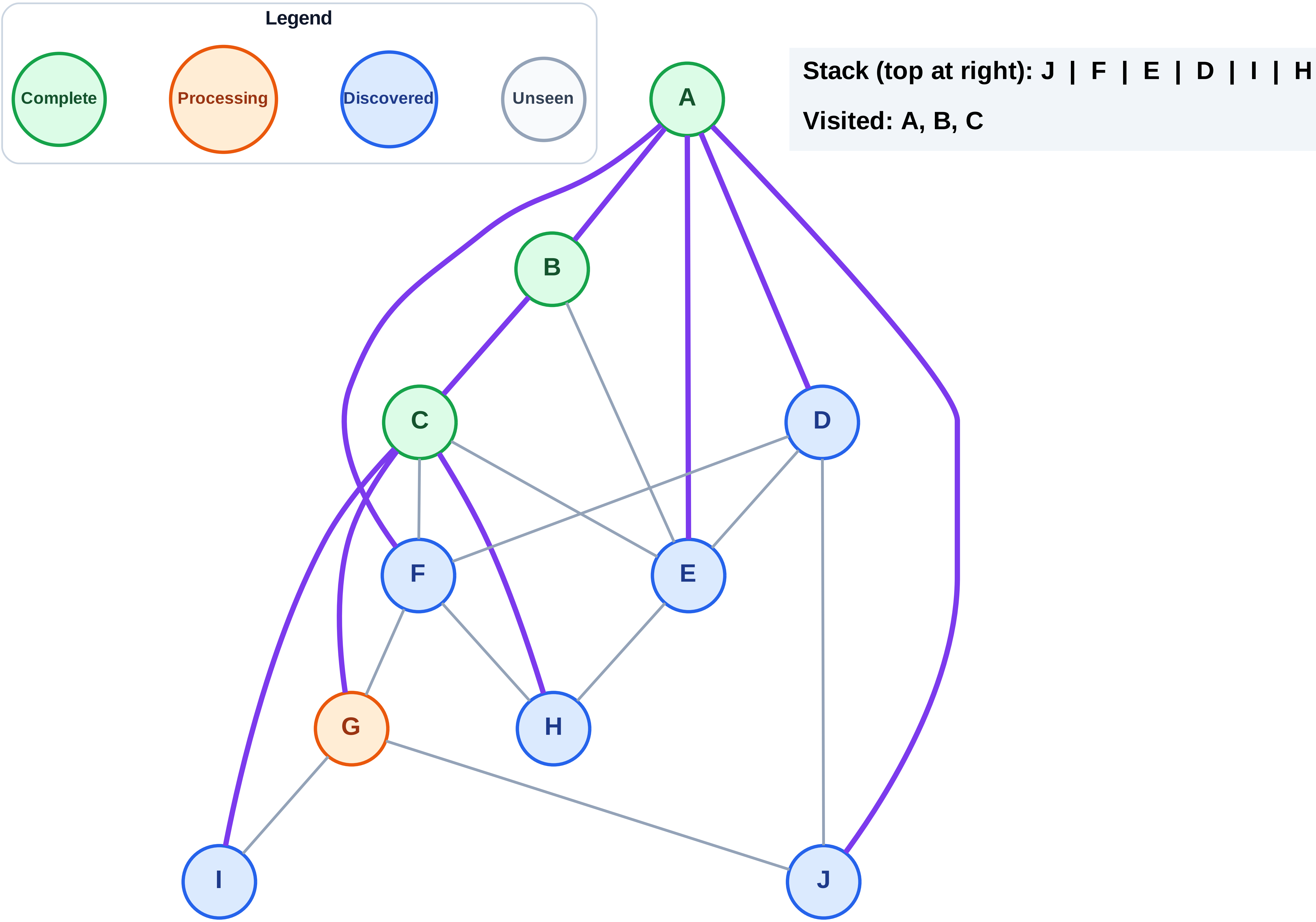
DFS Traversal

Step 7: Discover I, H, G from C



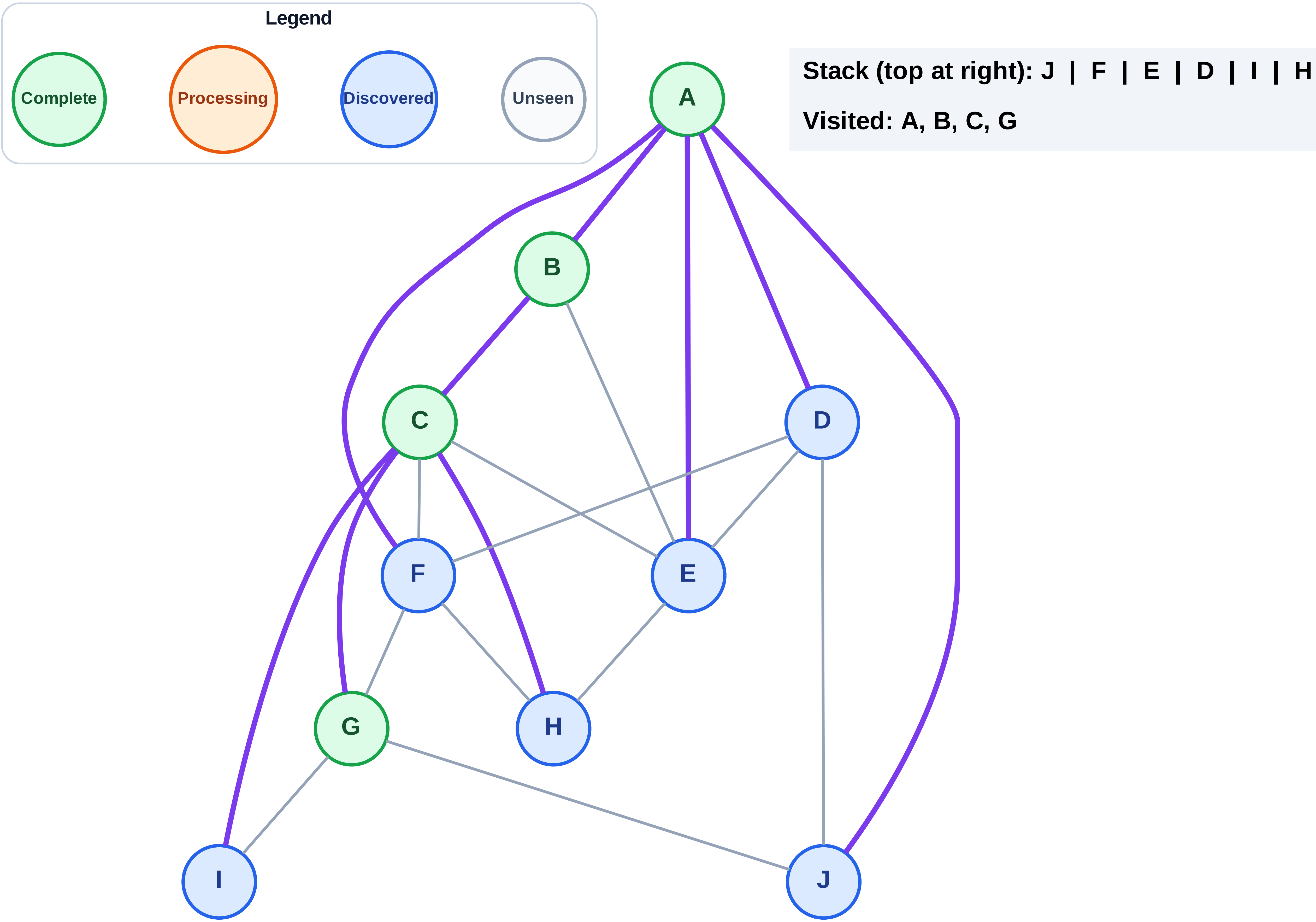
DFS Traversal

Step 8: Process node G



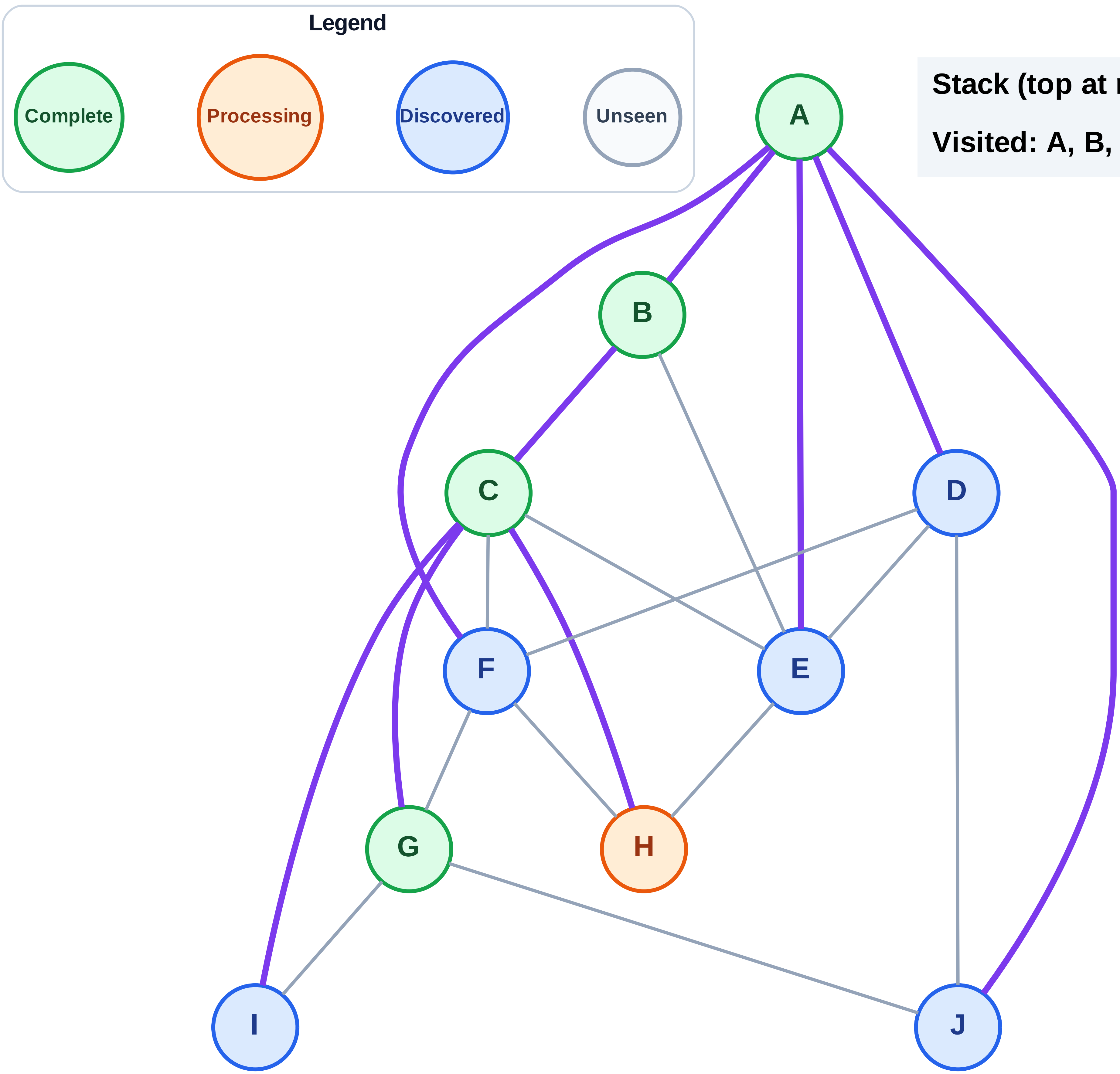
DFS Traversal

Step 9: No new neighbors from G



DFS Traversal

Step 10: Process node H

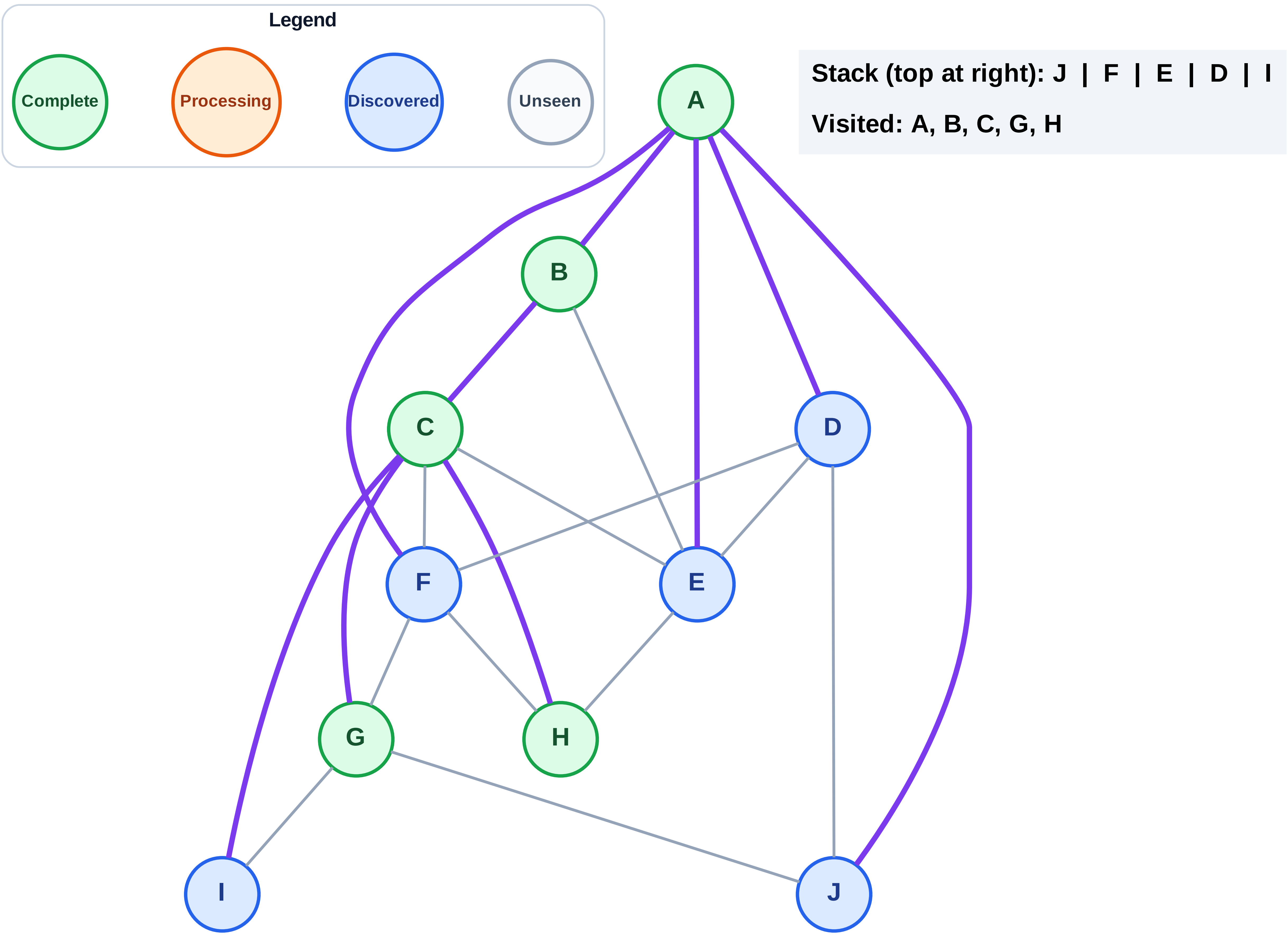


Stack (top at right): J | F | E | D | I

Visited: A, B, C, G

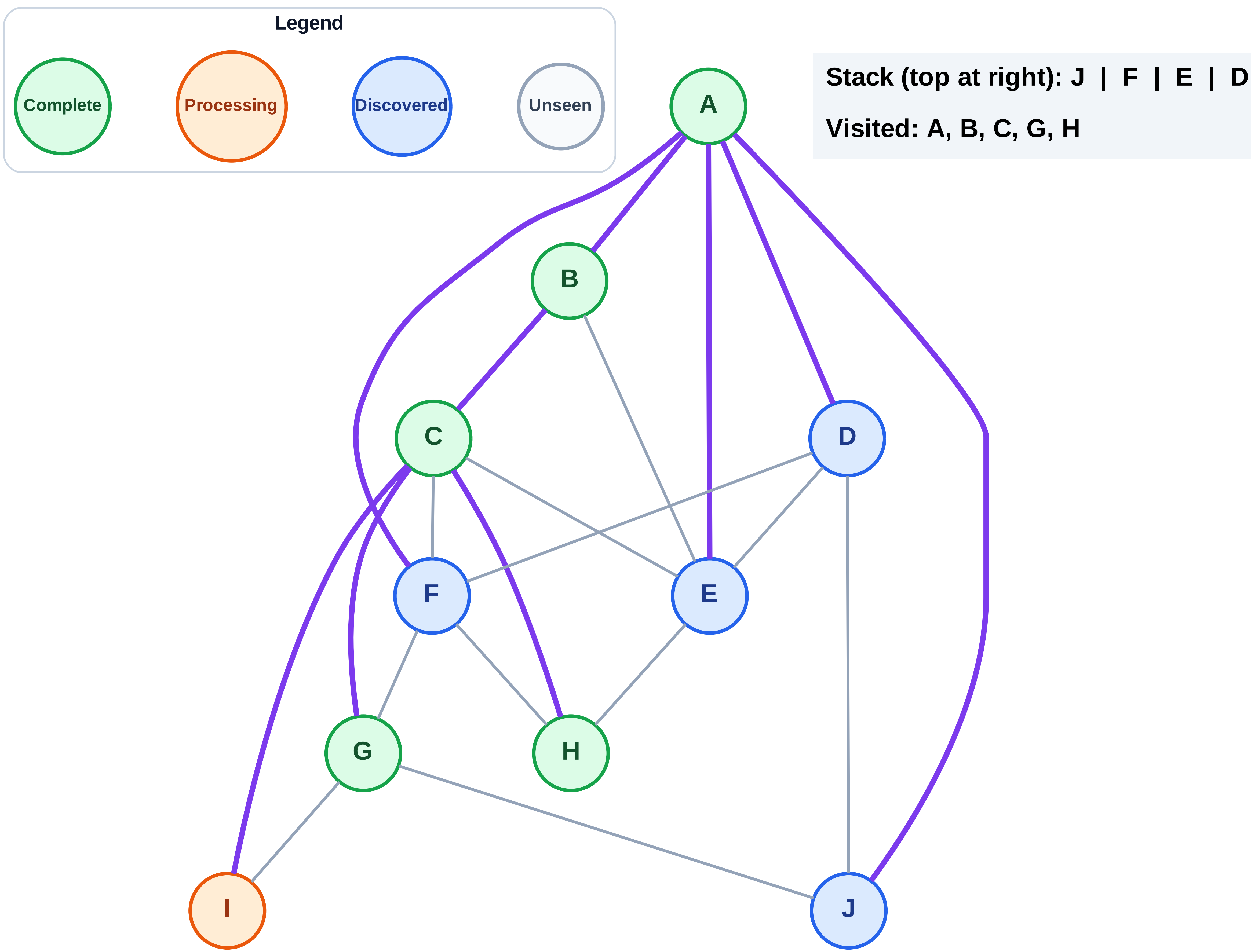
DFS Traversal

Step 11: No new neighbors from H



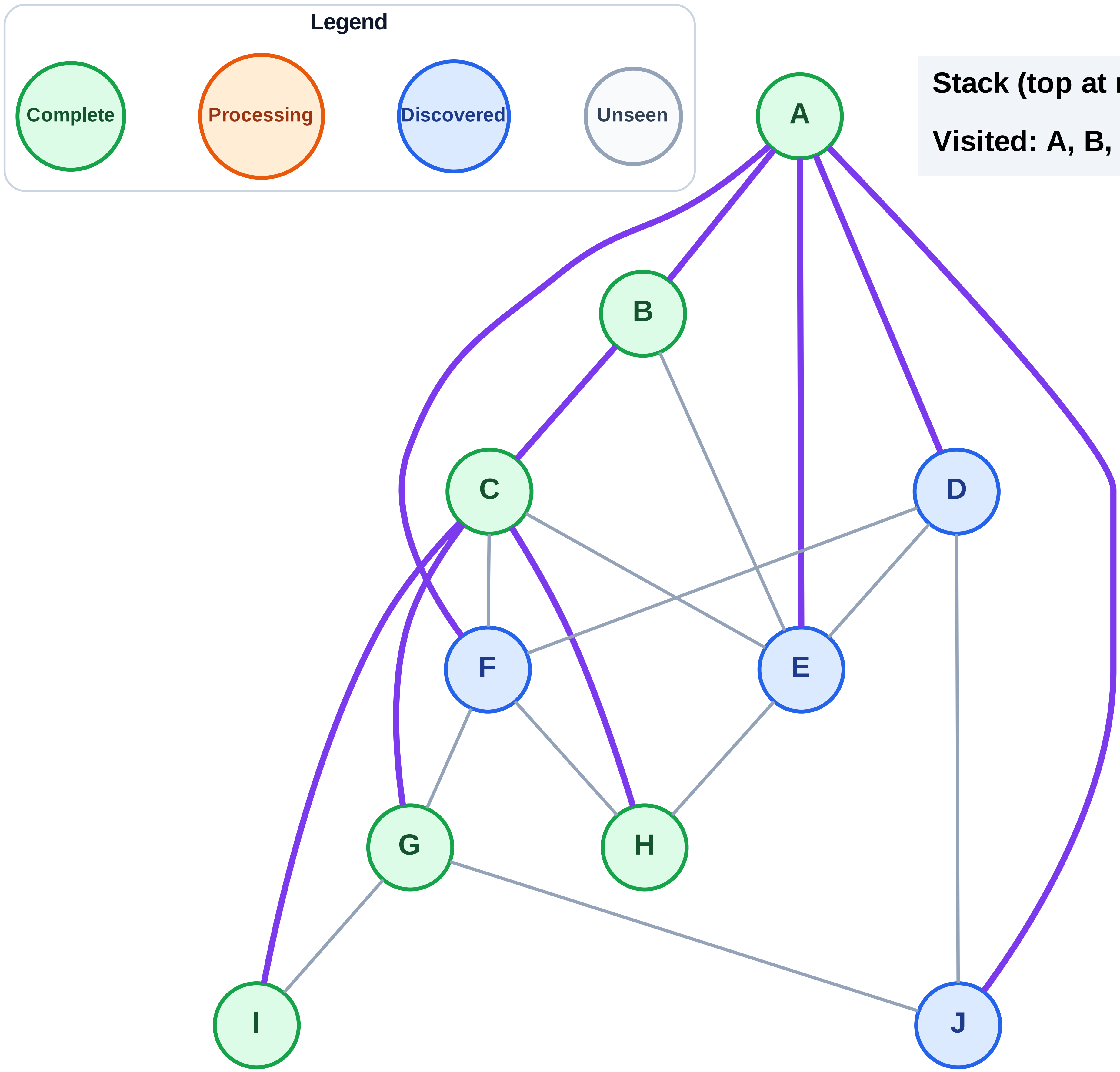
DFS Traversal

Step 12: Process node I



DFS Traversal

Step 13: No new neighbors from I

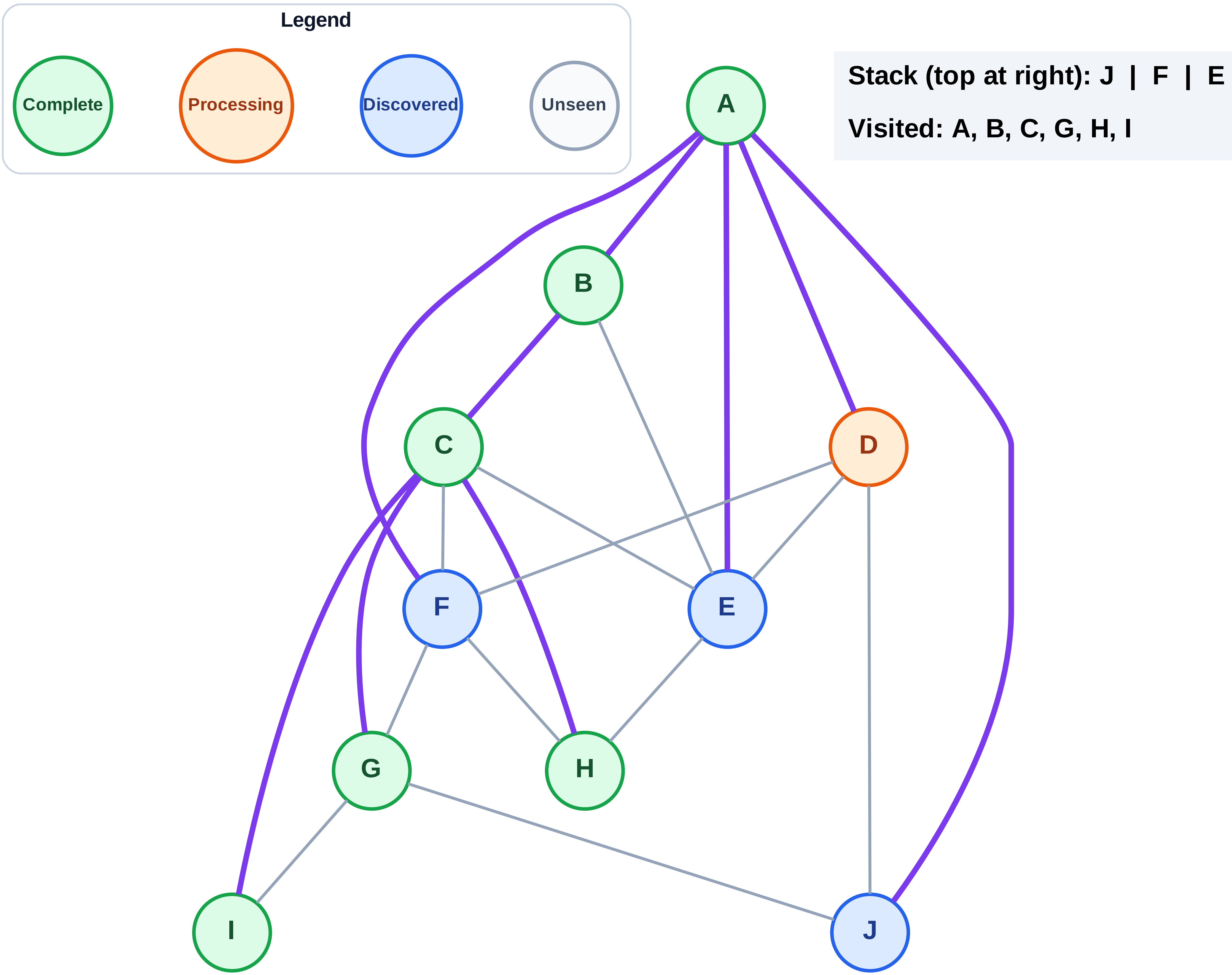


Stack (top at right): J | F | E | D

Visited: A, B, C, G, H, I

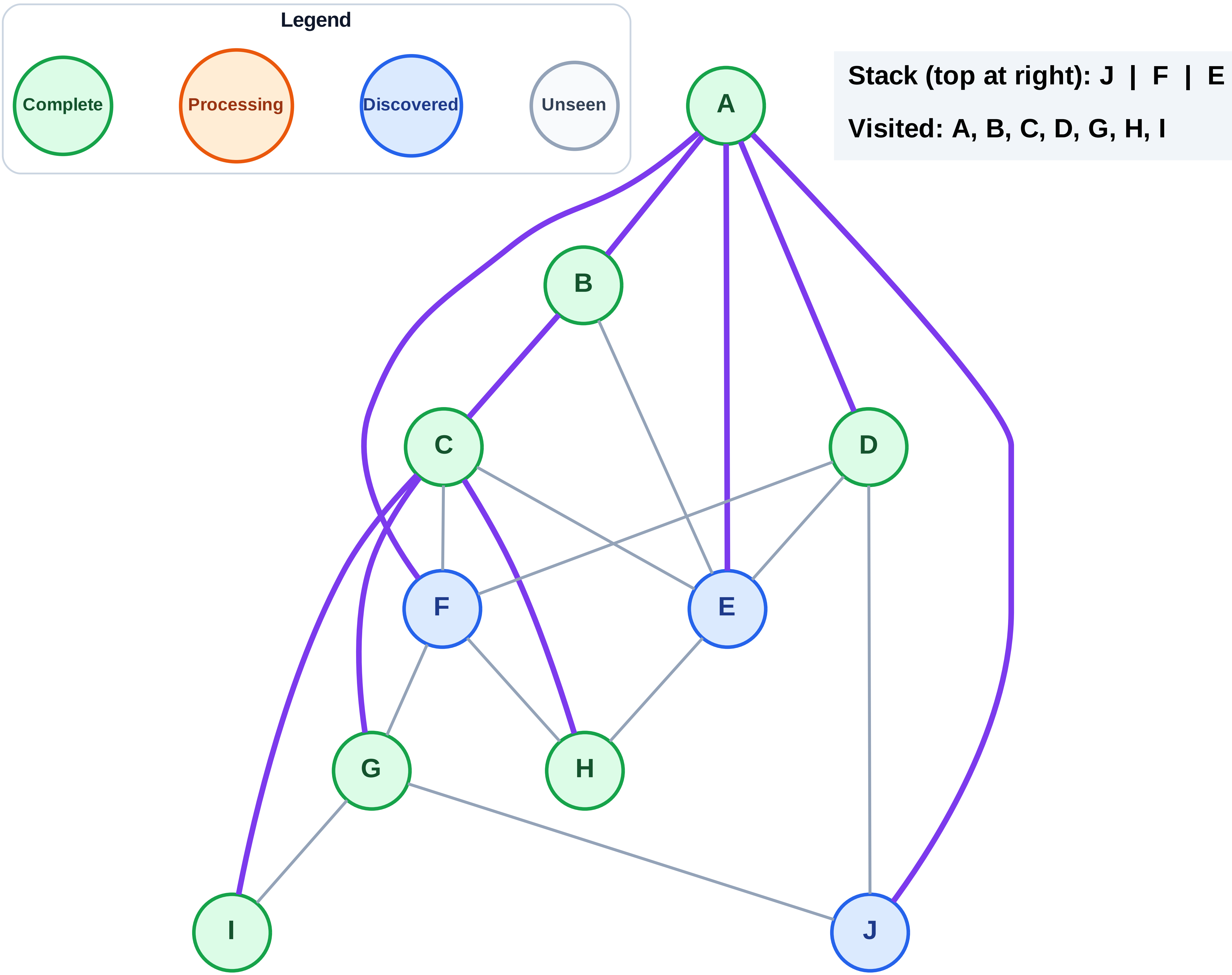
DFS Traversal

Step 14: Process node D



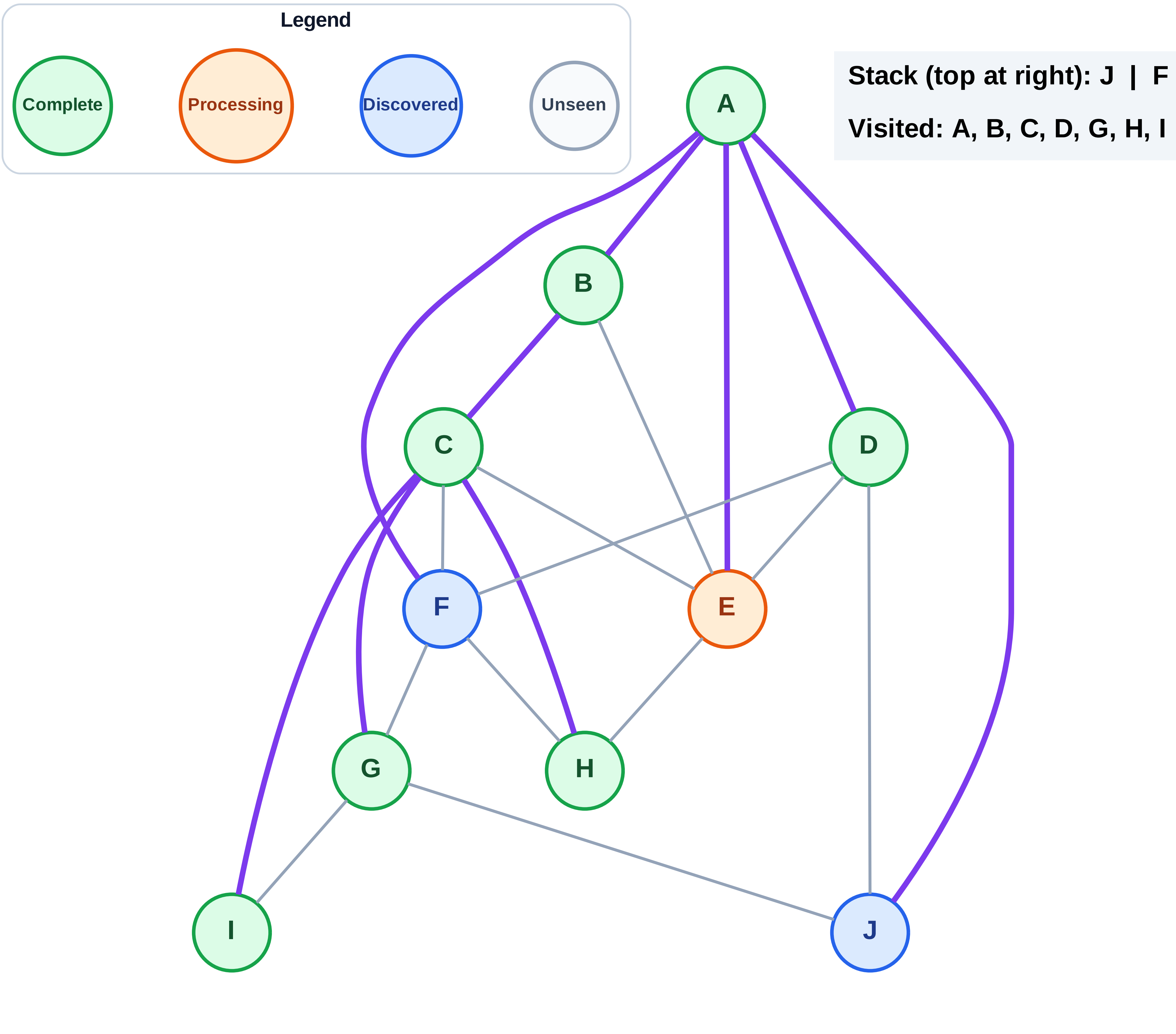
DFS Traversal

Step 15: No new neighbors from D



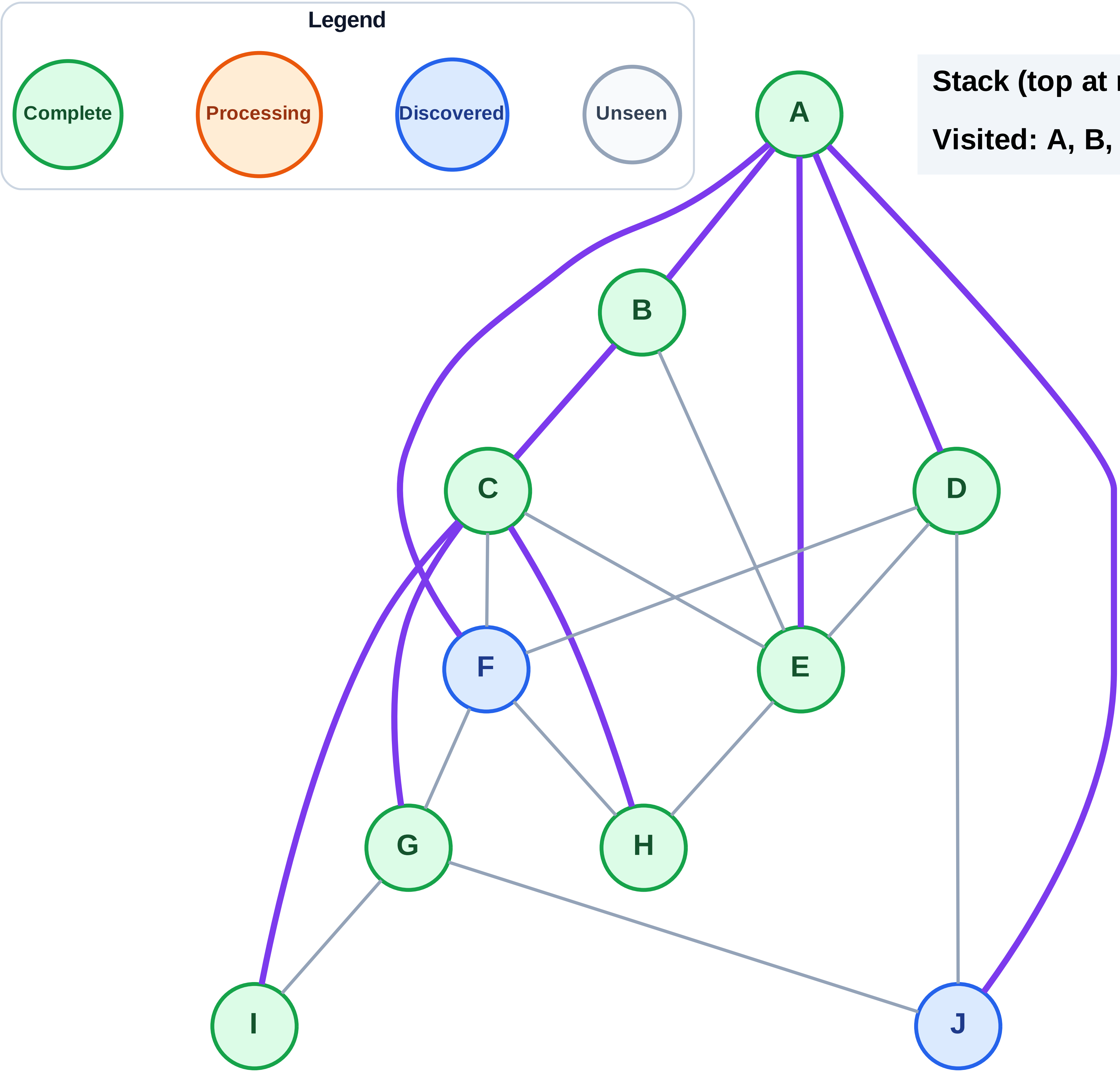
DFS Traversal

Step 16: Process node E



DFS Traversal

Step 17: No new neighbors from E

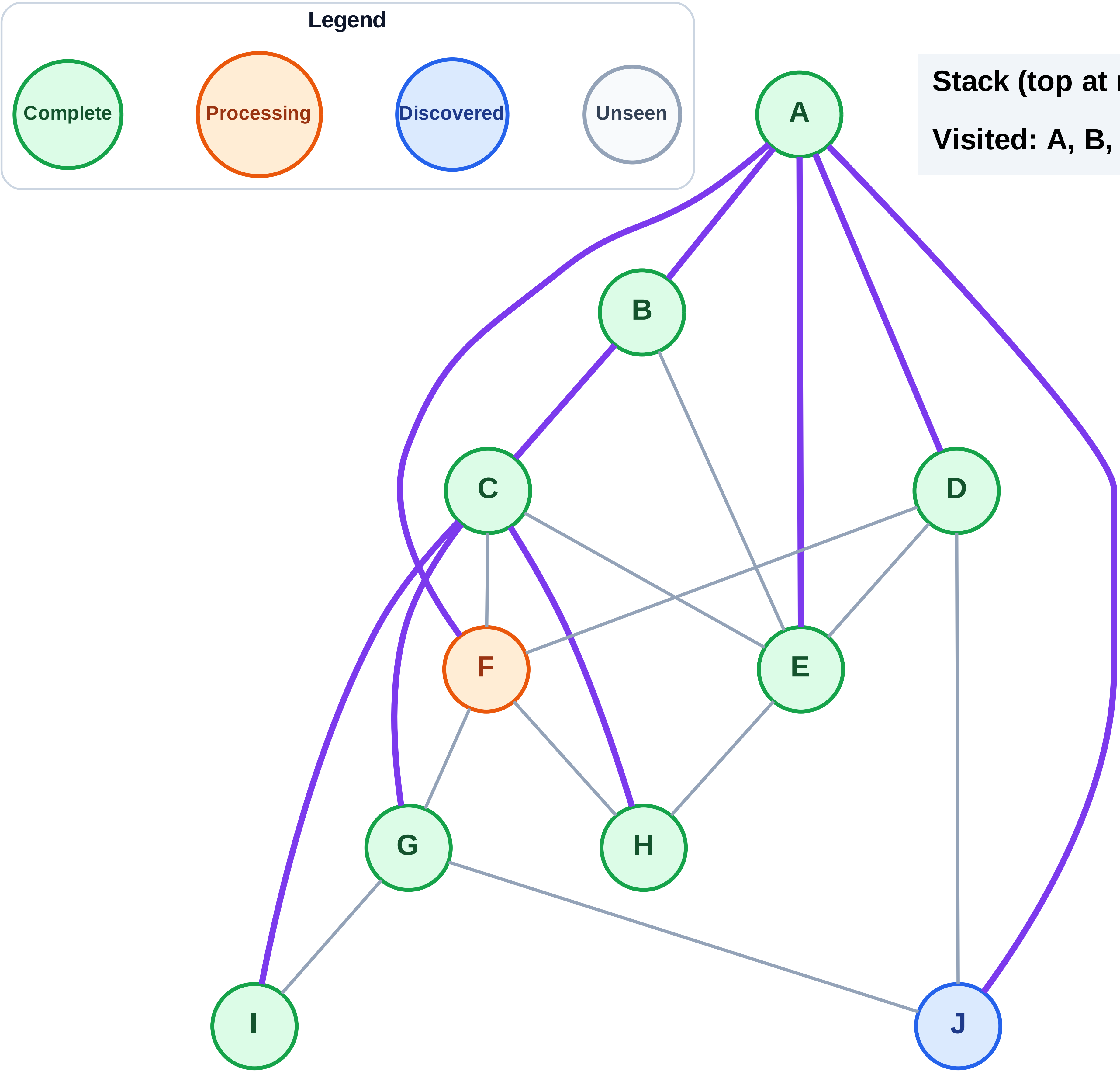


Stack (top at right): J | F

Visited: A, B, C, D, E, G, H, I

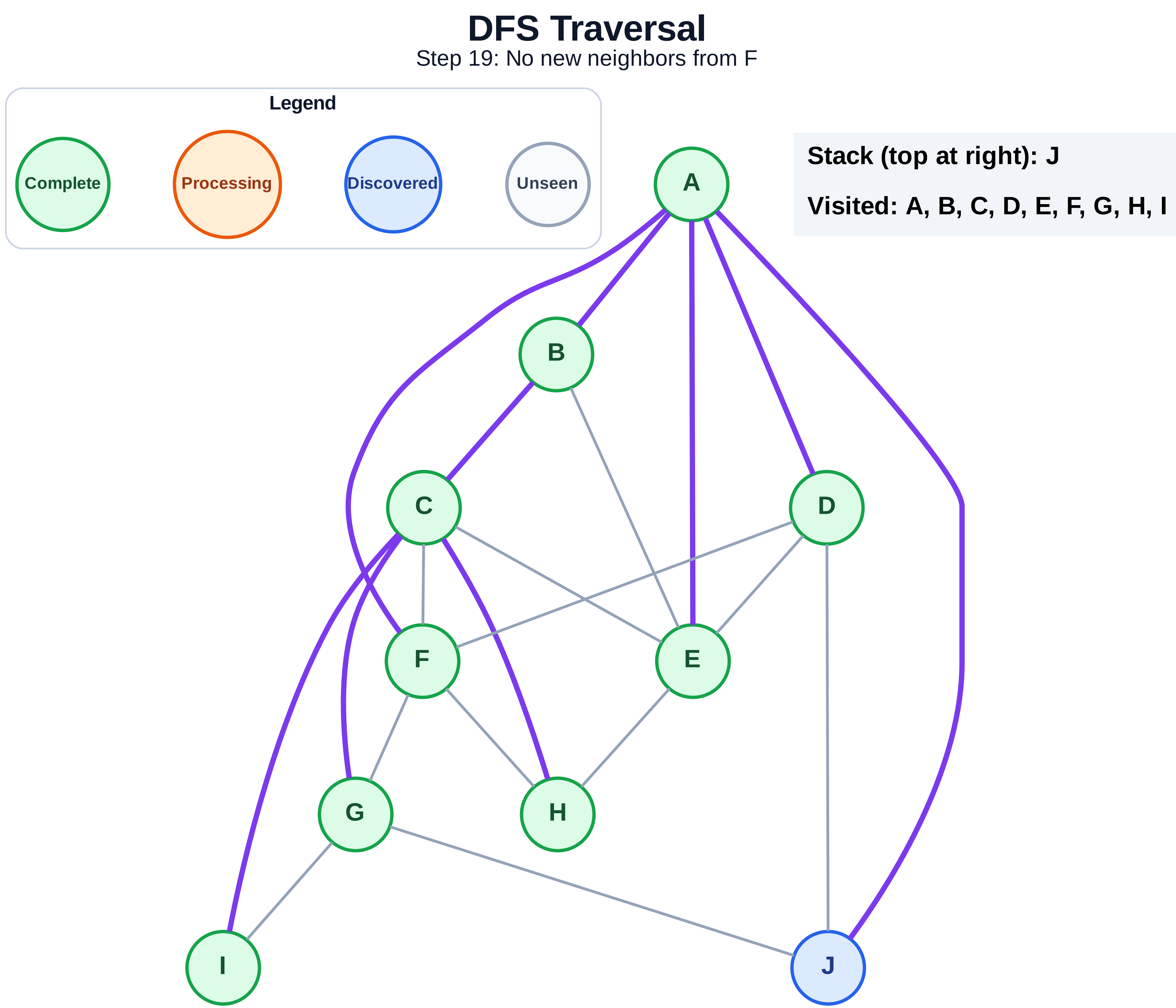
DFS Traversal

Step 18: Process node F

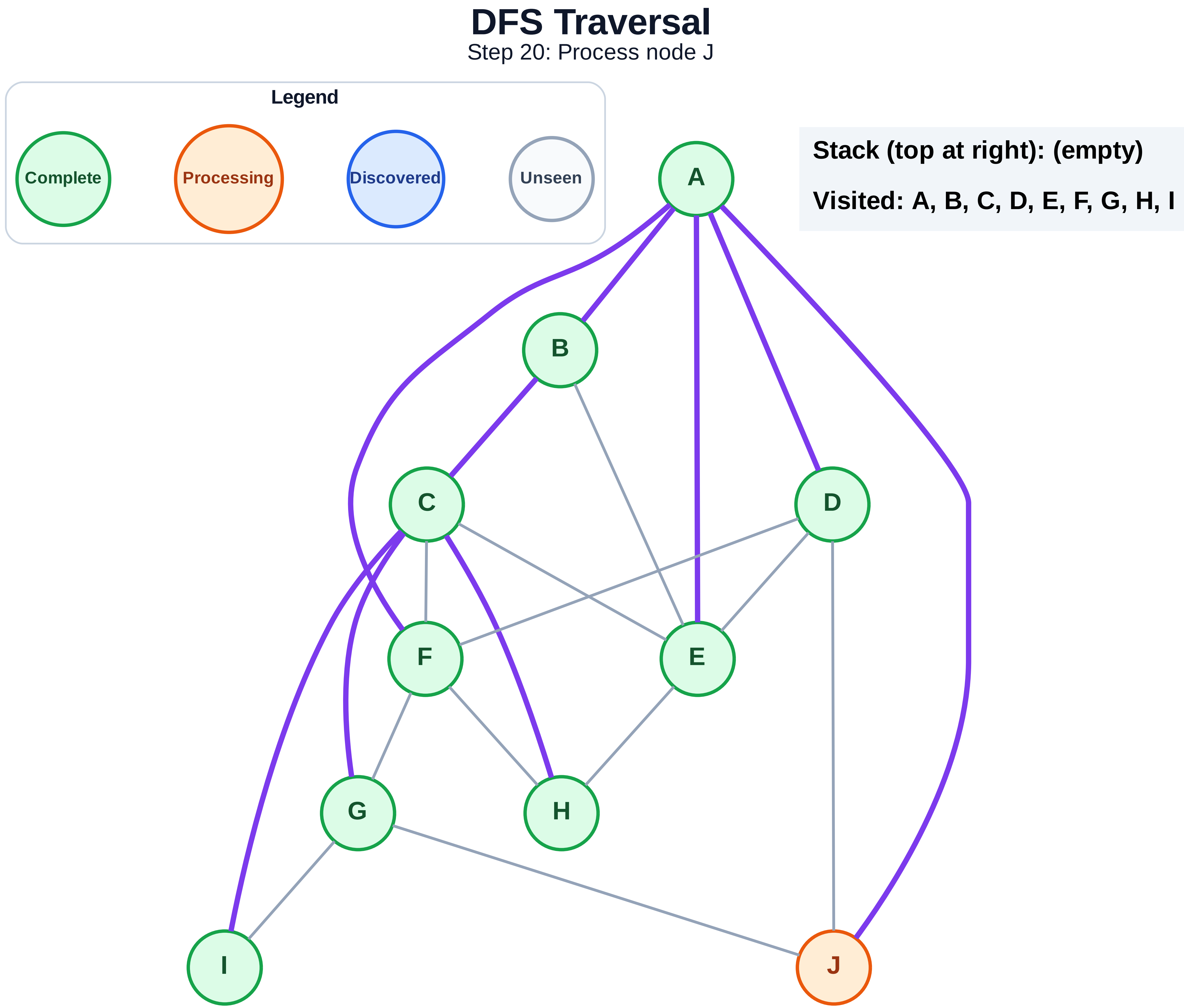


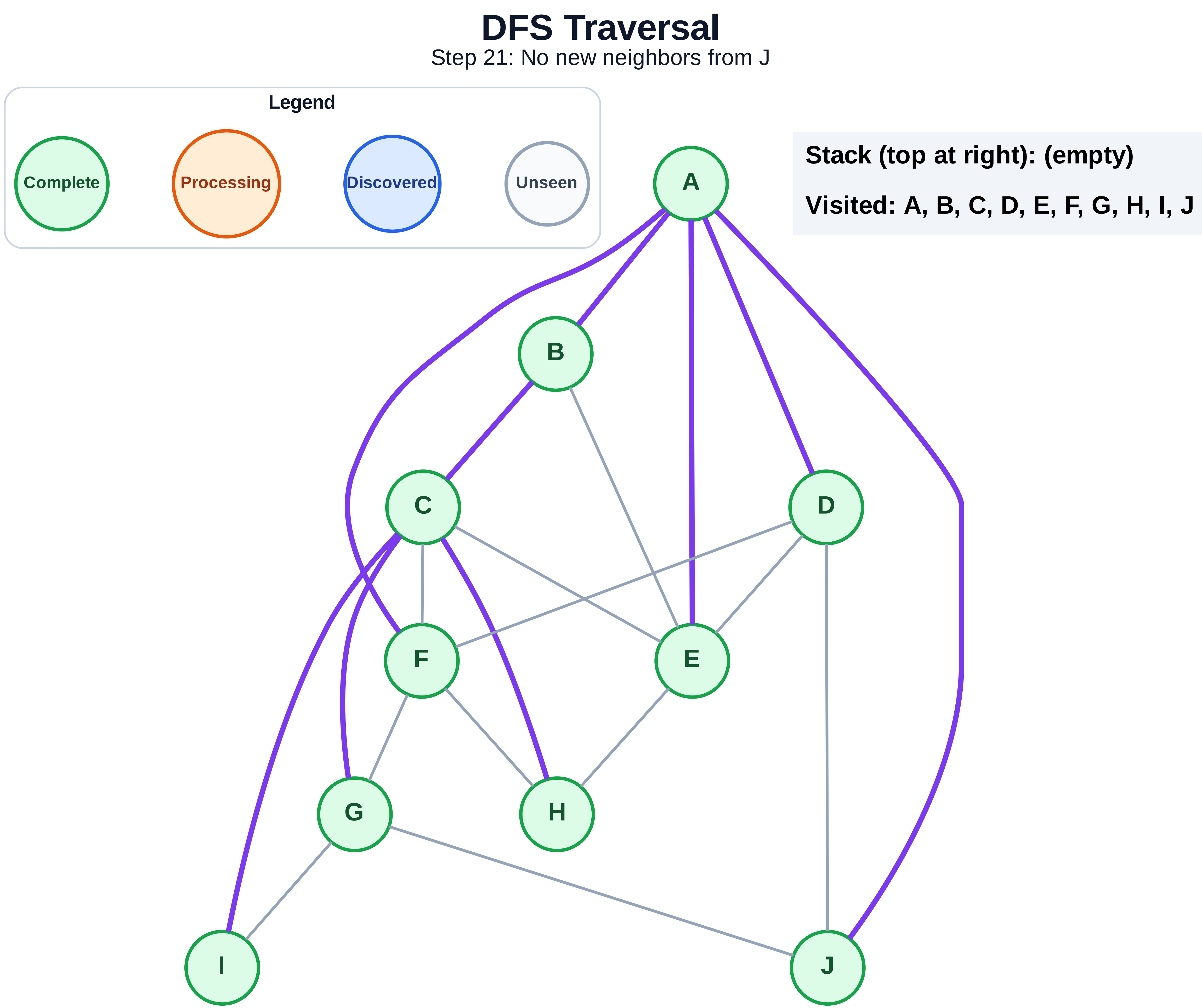
Stack (top at right): J

Visited: A, B, C, D, E, G, H, I



DFS Traversal
Step 19: No new neighbors from F
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): J
Visited: A, B, C, D, E, F, G, H, I
A
B
C
D
F
E
G
H
I
J


DFS Traversal
Step 20: Process node J
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): (empty)
Visited: A, B, C, D, E, F, G, H, I
A
B
C
D
E
F
G
H
I
J


DFS Traversal
Step 21: No new neighbors from J
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): (empty)
Visited: A, B, C, D, E, F, G, H, I, J
A
B
C
D
E
F
G
H
I
J